Babel

User guide

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Localization and internationalization

Unicode
TEX
LuaTEX
pdfTEX
XeTEX

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What is this document about? This user guide focuses on internationalization and localization with LaTeX and luatex, pdftex and xetex with the babel package. There are also some notes on its use with e-Plain and pdf-Plain TeX.

I only need learn the most basic features. The first subsections (1.1-1.6) describe the ways of loading a language, which is usually all you need.

I don't like manuals. I prefer sample files. This manual contains lots of examples and tips, but in GitHub there are many sample files.

What if I'm interested only in the latest changes? Changes and new features with relation to version 3.8 are highlighted with `X.XX` (※ is a link to the babel site), and there are some notes for the latest versions in the babel site. The most recent features can still be unstable. Remember version 24.1 follows 3.99, because of a new numbering scheme.

Can I help? Sure! You can follow the development of babel in GitHub and make suggestions, including requirements for some language or script. Feel free to fork it and make pull requests. If you are the author of a package, send me a few test files which I'll add to mine, so that possible issues can be caught in the development phase. Hyphenation rules are maintained separately here.

It doesn't work for me! You can ask for help in some forums like tex.stackexchange, but if you have found a bug, I strongly beg you to report it in GitHub, which is much better than just complaining on an e-mail list or a web forum. Remember *warnings are not errors* by themselves, they just warn about possible problems or incompatibilities.

How can I contribute a new language? See section 9.1 for contributing a language.

Where is the code? Run

```
lualatex --jobname=babel-code \let\babelcode\relax\input{babel.dtx}.
```

NOTE Now that the recommend engine for LaTeX is luatex, you can read Migrating from pdfTeX to LuaTeX and Migrating from XeTeX to LuaTeX.

1 The basic user interface

There are two ways to load a language with babel, namely, the old good 'classical' one, based on mostly self-contained declarations in a file with `ldf` extension, and the 'modern' one, based on a brand new mechanism based on descriptive `ini` files.

'Classical' doesn't mean outdated or obsolete. In fact, this is the recommended method in most languages where an `ldf` file exists. Below is a list of the supported languages. See also Which method for which language in the babel site.

Basically, what you need is typically:

- **Tell babel which language or languages are required.**
- **With non-Latin scripts and Unicode engines (luatex is the preferred one), select a suitable font (sec. 1.8)**
- **In multilingual documents, switch the language in the text body (sec. 1.9).**

You can find basic info and minimal luatex example files for about 300 locales in the GitHub repository. There are also some videos on Youtube.

1.1 Monolingual documents: the 'classical' way

In most cases, a single language is required, and then all you need in LaTeX is to load the package using its standard mechanism for this purpose, namely, passing that language as an optional argument. In addition, you may want to set the font and input encodings.

Another approach is making the language a global option in order to let other packages detect and use it. This is the standard way in LaTeX for an option – in this case a language –

to be recognized by several packages (in other words, babel doesn't set the languages, it just recognizes the options passed to the class or the package).

Many languages are compatible with luatex and xetex, but a few only work with pdftex. When these engines are used, the Latin script is covered by default in current LaTeX (provided the document encoding is UTF-8). Other scripts require loading fontspec, although babel provides a higher level interface (see `\babelfont` below).

25.8 ※ The PDF document language can also be set with `\DocumentMetadata` before `\documentclass`. See the liked news page for further info. For example, the following setting will pass `british` as the main language to babel:

```
\DocumentMetadata{language=en-GB}
\documentclass{article}
\usepackage{babel}
```

26.9 ※ The option `other-languages` is also recognized.

The basic tag lookup explained below is applied here, so that `fr-Latn-FR` is valid and mapped to `fr`, which is in turn mapped to `french`. This eases the localization of automatically generated documents.

WARNING This is a **breaking change**, because until now with this example the dummy language `nil` was loaded. Tags are not validated, which means mistakes like `en-UK` or wrong tags like `de-AUS` or `la-x-classical` might be accepted without complaining.

EXAMPLE Here is a simple full example for "traditional" TeX engines (see below for luatex and xetex). The package `fontenc` does not belong to babel, but it is included in the example because typically you will need it. It assumes UTF-8, the default encoding:

```
\documentclass{article}

\usepackage[T1]{fontenc}

\usepackage[french]{babel}

\begin{document}

Plus ça change, plus c'est la même chose!

\end{document}
```

Now consider something like:

```
\documentclass[french]{article}
\usepackage{babel}
\usepackage{varioref}
```

With this setting, the package `varioref` will also see the option `french` and will be able to use it.

EXAMPLE Now a simple monolingual document in Russian (text from the Wikipedia) with luatex or xetex. Note neither fontenc nor inputenc is necessary, and a so-called Unicode font must be loaded (in this example with the help of `\babelfont`, described below).

```
\documentclass[russian]{article}

\usepackage{babel}

\babelfont{rm}{DejaVu Serif}

\begin{document}

Россия, находящаяся на пересечении множества культур, а также
```

```
с учётом многонационального характера её населения, — отличается
высокой степенью этнокультурного многообразия и способностью к
межкультурному диалогу.

\end{document}
```

NOTE Because of the way babel has evolved, “language” can refer to (1) a set of hyphenation patterns as preloaded into the format, (2) a package option, (3) an `ldf` file, and (4) a name used in the document to select a language. Please, read the documentation for specific languages for further info.

NOTE Babel does not make any readjustments by default in font size, vertical positioning or line height. This is on purpose because the optimal solution depends on the document layout and the font, and very likely the most appropriate one is a combination of these settings. A quick and simple solution to get a more uniform vertical spacing is with the option `12pt` in standard classes, and then reducing the font with `Scale` (memoir provides also larger values, like `14pt`).

NOTE Although it has been customary to recommend placing `\title`, `\author` and other elements printed by `\maketitle` after `\begin{document}`, mainly because of shorthands, it is advisable to keep them in the preamble. Currently there is no real need to use shorthands in those macros (and if you need them, you can use `\babelshorthand`).

NOTE With hyperref you may want to set the PDF document language with something like:

```
\usepackage[pdflang=es-MX]{hyperref}
```

This is not currently done by babel and you must set it by hand.

WARNING In the preamble, *no* language has been selected, except hyphenation patterns and the name assigned to `\localename` (and `\languagename`). In particular, shorthands, captions and date are *not* activated. If you need to define boxes and the like in the preamble, you might want to use some of the language selectors described below.

TROUBLESHOOTING *Package inputenc Error: Invalid UTF-8 byte ...* A common source of trouble is a wrong setting of the input encoding. Make sure you set the encoding actually used by your editor, or even better, make sure the encoding in your editor is set to UTF-8.

TROUBLESHOOTING Another typical error when using babel is the following:

```
! Package babel Error: Unknown option 'LANG'.
(babel)                Suggested actions:
(babel)                * Make sure you haven't misspelled it
(babel)                * Check in the babel manual that it's supported
(babel)                * If supported and it's a language, you may
(babel)                  need in some distributions a separate
(babel)                  installation
(babel)                * If installed, check there isn't an old
(babel)                  version of the required files in your system
(babel)                * If it's an unsupported language, create it with
(babel)                  \babelprovide (see the manual).
```

Sometimes, the error vanishes when the document is typeset again, but in more severe ones you will need to remove the `aux` file.

TROUBLESHOOTING *No hyphenation patterns were preloaded...* The following warning is about hyphenation patterns, which are not under the direct control of babel:

```
Package babel Warning: No hyphenation patterns were preloaded for
(babel)                the language `LANG' into the format.
(babel)                Please, configure your TeX system to add them and
(babel)                rebuild the format. Now I will use the patterns
(babel)                preloaded for \language=0 instead on input line 57.
```

The document will be typeset, but very likely the text will not be correctly hyphenated. Some languages in some system may be raising this warning wrongly (because they are not hyphenated) – just ignore it. See the manual of your distribution (MacTEX, MikTEX, TEXLive, etc.) for further info about how to configure it.

TROUBLESHOOTING *You are loading directly a language style.* Loading directly `sty` files in LaTeX (i.e., `\usepackage{`⟨*language*⟩`}`) was deprecated many years ago and you will get the error:

```
! Package babel Error: You are loading directly a language style.
(babel)                This syntax is deprecated and you must use
(babel)                \usepackage[language]{babel}.
```

NOTE You will see an 'info' in the log file stating some data is being loaded from an `ini` file. It includes standardized names for language and script used by `\babelfont`, and the BCP 47 tag, required in some cases by `\Make`⟨*Xxxxx*⟩`case`.

1.2 Monolingual documents: the 'modern' way

WARNING 25.14 ※ Languages not supported with the `ldf` mechanism are new loaded without `provide=*`. This manual has not been fully updated yet. There is also a new value: `provide=!`, which forces the ldf file (for example, for local styles in a publishing house or if a new style is being developed).

When, for whatever reason, the 'classical' way with the `ldf` is not suitable for the needs of a document or a document system, you have to resort to a different mechanism, which is activated automatically if the language is not supported by the 'traditional' way or with the package option `provide=*` (in monolingual documents).

EXAMPLE Although Georgian has its own `ldf` file, here is how to declare this language in Unicode engines with this mechanism. Here, as in a previous example, we resort to `\babelfont` to set the font for this language (with the `Harfbuzz` renderer, just to show how to force it, because here the `Node` renderer should be fine).

```
\documentclass{book}

\usepackage[georgian, provide=*]{babel}

\babelfont{rm}{DejaVu Sans}[Renderer=Harfbuzz]

\begin{document}

\tableofcontents

\chapter{სამზარეულო და სუფრის ტრადიციები}

ქართული ტრადიციული სამზარეულო ერთ-ერთი უმდიდრესია მთელ მსოფლიოში.

\end{document}
```

And with a global option:

```
\documentclass[georgian]{book}
\usepackage[provide=*]{babel}
```

```
\babelfont{rm}{DejaVu Sans}[Renderer=Harfbuzz]
```

NOTE The option `provide` actually loads the language with `\babelprovide` and the `import` option, described below. Instead of an asterisk, you may provide a list of options for `\babelprovide` enclosed in braces (`import` is included by default).

EXAMPLE For a text in Chinese, you can write in the preamble:

```
\usepackage[chinese]{babel}
\babelfont{rm}{FandolSong}
```

The default skip between characters is 0 em plus .1 em, which can be modified with an option (explained below) in `provide`. For example:

```
\usepackage[chinese, provide={ intraspace=0 .2 0 }]{babel}
```

WARNING Because of the way LaTeX passes options, with `provide` there must be *no* space in the middle of `={`, as the previous example shows.

1.3 Mostly monolingual documents: lazy loading

3.39 ※ Very often, multilingual documents consist of a main language with small pieces of text in another languages (words, idioms, short sentences). Typically, all you need is to set the line breaking rules and, perhaps, the font. In such a case, there is no need to clutter the preamble (particularly the class and package options) because babel does not require declaring these secondary languages explicitly — the basic settings are loaded on the fly when the language is first selected in the document.

This is particularly useful in documents with many languages, and also when there are short texts of this kind coming from an external source whose contents are not known beforehand (for example, titles in a bibliography). In this regard, it is worth remembering that `\babelfont` does *not* load any font until required, so that it can be used just in case.

3.84 ※ With pdftex, when a language is loaded on the fly (internally it's loaded with `\babelprovide`) selectors now set the font encoding based on the list provided when loading `fontenc`. Not all scripts have an associated encoding, so this feature works only with Latin, Cyrillic, Greek, Arabic, Hebrew, Cherokee, Armenian, and Georgian, provided a suitable font is found.

EXAMPLE A trivial document with the default font in English and Spanish, and FreeSerif in Russian is:

```
\documentclass[english]{article}
\usepackage{babel}

\babelfont[russian]{rm}{FreeSerif}

\begin{document}

English. \foreignlanguage{russian}{Русский}.
\foreignlanguage{spanish}{Español}.

\end{document}
```

If you need the 'modern' way to load the main language set as global option, just write:

```
\usepackage[provide=*]{babel}
```

NOTE Instead of its name, you may prefer to select the language with the corresponding BCP 47 tag. This alternative, however, must be activated explicitly, because a two- or three-letter word is a valid name for a language (e.g., `lu` can be the locale name with tag `khb` or the tag for `lubakatanga`). See section 2.6 for further details.

1.4 Multilingual documents: the 'classical' way

In multilingual documents, just use a list of the required languages either as package or as class options. The last language is considered the main one, activated by default. Sometimes, the main language changes the document layout (e.g., `spanish` and `french`).

To switch the language there are two basic macros, described below in detail: `\selectlanguage` is used for blocks of text, while `\foreignlanguage` is for chunks of text inside paragraphs.

EXAMPLE In LaTeX, the preamble of the document:

```
\documentclass{article}
\usepackage[dutch,english]{babel}
```

would tell LaTeX that the document would be written in two languages, Dutch and English, and that English would be the first language in use, and the main one.

EXAMPLE A full bilingual document with pdftex follows. The main language is `french`, which is activated when the document begins. It assumes UTF-8:

```
\documentclass{article}

\usepackage[T1]{fontenc}

\usepackage[english,french]{babel}

\begin{document}

Plus ça change, plus c'est la même chose!

\selectlanguage{english}

And an English paragraph, with a short text in
\foreignlanguage{french}{français}.

\end{document}
```

EXAMPLE With luatex and xetex, the following bilingual, single script document in UTF-8 encoding just prints a couple of 'captions' and `\today` in Danish and Vietnamese. No additional packages are required, because the default font supports both languages.

```
\documentclass{article}

\usepackage[vietnamese,danish]{babel}

\begin{document}

Danish: \prefacename, \alsoname, \today.

\selectlanguage{vietnamese}

Vietnamese: \prefacename, \alsoname, \today.

\end{document}
```

NOTE Although strongly discouraged, languages can be set as global and as package option at the same time, but in such a case you should set explicitly the main language with the package option `main`, described below:

```
\documentclass[italian]{book}
\usepackage[german,main=italian]{babel}
```

NOTE Once a language is loaded, you can select it with the corresponding BCP 47 tag. See section 2.6 for further details.

NOTE (pdftex) Documents with several input encodings are not frequent, but sometimes are useful. You can set different encodings for different languages as the following example shows:

```
\AddBabelHook{encodings}{afterextras}{\inputencoding{latin1}}
\AddBabelHook[russian]{encodings}{afterextras}{\inputencoding{koi8-r}}
```

See the description of `\AddBabelHook` below. The first line activates the `latin1` input encoding in every language. The second one, which overrides the previous one, activates `koi8-r` in Russian.

1.5 Multilingual documents: the 'modern' way

If lazy loading is not enough for your purposes, you can still tell which languages should be loaded as either class or package options. You can combine the 'classical' and the 'modern' ways with three options to set which method you want, which cover the most typical cases [to be revised; a new 'value' `!` forces the `ldf` mode for the main language]:

- `provide=*` is the option explained above for monolingual documents. If there are more languages, it applies only to the main language, while the rest will be loaded in the 'classical' way..
- `provide+=*` loads the main language in the 'classical' way, and the rest in the 'modern' one.
- `provide*=*` is the same for all languages, i.e., main and secondary ones are loaded in the 'modern' way.

More complex combinations can be handled with `\babelprovide`, explained below.

EXAMPLE Your document is written in Thai with large chunks in Dutch and German, and you want the `ldf` files in the latter because, for example, you need their shorthands. The font is Norasi, which covers the three languages:

```
\usepackage[dutch,german,thai]{babel}
\babelfont{rm}{Norasi}
```

This will load `dutch` and `german` in the classical `ldf` mode, but `thai` in the modern `ini` mode. Other options are:

```
  \usepackage[dutch,german,thai,provide=!]{babel}
```

which will use the classical mode for in languages (as described in the previous section; `provide=!` forces the `ldf` file, but note `thai.ldf` is not supported in Unicode engines), and:

```
  \usepackage[dutch,german,thai,provide*=*]{babel}
```

which will use the modern mode in all languages.

1.6 Languages supported by babel in the 'classical' mode

(To be updated.) In the following table most of the languages supported by babel with an `ldf` file are listed, together with the names of the options which you can load babel with for each language. Note this list is open and the current options may be different. It does not include `ini` files (see below).

Except in a few cases (e.g., `serbianc`, `acadian`) names are those of the Unicode CLDR (or based on them; see *Locale naming*).

Most of them work out of the box, but some may require extra fonts, encoding files, a preprocessor or even a complete framework (like CJK or luatexja).

This list is still under revision.
Recommended names are set in red.
Additional languages are set in gray.
Discouraged and deprecated names are not included.
There are some notes in a few languages.
The reference mark (※) is a link to CTAN.

---

afrikaans ※
albanian ※
amharic ※
  Calls ethiop
arabic ※
  Calls arabi.
azerbaijani ※
basque ※
belarusian ※
bosnian ※
breton ※
bulgarian ※
catalan ※
croatian ※
czech ※
danish ※
dutch ※
english ※
  american
  USenglish
  australian
  british
  UKenglish
  canadian
  newzealand
esperanto ※
estonian ※
ethiop
  See amharic.
farsi ※
  Calls arabi.
finnish ※
french ※
  acadian
friulian ※
galician ※
georgian ※
  Calls geotex.
german ※
  See the manual for the german babel style.
  austrian
greek ※
  polytonicgreek
    Calls greek.polytonic
  ancientgreek
    Calls greek.ancient
  ibycus
  bgreek
hebrew ※
hindi ※
  Calls velthuis.
hungarian ※
  magyar
icelandic ※
indonesian ※
interlingua ※
irish ※
italian ※
japanese ※
kurmanji
latin ※
  classicallatin
  medievallatin
  ecclesiasticallatin
latvian ※
lithuanian ※
lowersorbian ※
macedonian ※
malay ※
mongolian ※
ngerman
  See the manual for the german babel style.
  naustrian
  nswissgerman
    Swiss High German
northernsami ※
norwegian ※
  norsk
nynorsk ※
occitan ※
piedmontese ※
pinyin
polish ※
portuguese ※
  brazilian
romanian ※
romansh ※
russian ※
scottishgaelic ※
  scottish
serbianc ※
    Cyrillic script
  serbian
    Latin script
slovak ※
slovenian ※
spanglish ※
spanish ※
swedish ※
thai ※
  thaicjk
turkish ※
turkmen ※

ukrainian ※
uppersorbian ※
vietnamese ※
welsh ※

EXAMPLE (pdftex) An example of a language requiring a preprocessor and a separate package is `hindi`. If you have got the velthuis/devnag package, create a file with extension `.dn`:

```
\documentclass{article}
\usepackage[hindi]{babel}
\begin{document}
{\dn devaanaa.m priya.h}
\end{document}
```

Then preprocess it with `devnag` ⟨*file*⟩, which creates ⟨*file*⟩`.tex`; you can then typeset the latter with LaTeX.

1.7 Languages supported by babel in the 'modern' mode

Here is the list of the names currently supported with `ini` locale files, with `\babelprovide` (or `provide=`), or loaded by default when there is no `ldf` file. With these languages, `\babelfont` loads (if not done before) the language and script names (even if the language is defined as a package option with an ldf file). These are also the names recognized by `\babelprovide` with a valueless `import`, which will load the `ini` file with the tag given in parenthesis.

Following the current common practice (for example, the Unicode CLDR), all locales are organized in a flat structure. This eases their identification and customization.

Many locales are quite usable, provided captions and dates are not required (which is a very frequent case, particularly in ancient languages). So, they are included in the default babel distribution. This can serve to encourage contributions, too. A warning will remember they are 'bare minimum locales'. They are set in gray in the following list.

NOTE Wikipedia defines a *locale* as follows: "In computing, a locale is a set of parameters that defines the user's language, region and any special variant preferences that the user wants to see in their user interface. Usually a locale identifier consists of at least a language code and a country/region code." Babel is moving gradually from the old and fuzzy concept of *language* to the more modern of *locale*. Note each locale is by itself a separate "language", which explains why there are so many files. This is on purpose, so that possible variants can be created and/or redefined easily.

Recommended names are set in red.
In variants with the region or the script name (which are not highlighted), prefer the full forms.
Bare minimum locales are set in gray.
Discouraged and deprecated names are not included.
[u] means Unicode captions; [l] means LICR captions.
There are some notes in a few locales.
The reference mark (※) is a link to the babel site.

---

abkhazian (ab) ※
afar (aa) ※
afrikaans[ul] (af) ※
aghem (agq) ※
akan (ak) ※
akkadian (akk) ※
albanian[ul] (sq) ※
algerianarabic (arq) ※
Darija or Lahja Djazairia, with tag `arq`, different from Standard Arabic as spoken in Alger, with tag `ar-DZ`.
amharic[ul] (am) ※
ancientegyptian (egy) ※
ancientgreek[ul] (grc) ※
It's a different language from `greek`.
ancienthebrew (hbo) ※
arabic[u] (ar) ※
arabic-algeria[u] (ar-DZ)

- arabic-dz[u] (ar-DZ)
- arabic-egypt[u] (ar-EG)
- arabic-eg[u] (ar-EG)
- arabic-iraq[u] (ar-IQ)
- arabic-iq[u] (ar-IQ)
- arabic-jordan[u] (ar-JO)
- arabic-jo[u] (ar-JO)
- arabic-lebanon[u] (ar-LB)
- arabic-lb[u] (ar-LB)
- arabic-morocco[u] (ar-MA)
- arabic-ma[u] (ar-MA)
- arabic-palestinianterritories[u] (ar-PS)
- arabic-ps[u] (ar-PS)
- arabic-saudiarabia[u] (ar-SA)
- arabic-sa[u] (ar-SA)
- arabic-syria[u] (ar-SY)
- arabic-sy[u] (ar-SY)
- arabic-tunisia[u] (ar-TN)
- arabic-tn[u] (ar-TN)

aramaic (arc) ※
- aramaic-nabataean (arc-nbat)
- aramaic-nbat (arc-nbat)
- aramaic-palmyrene (arc-palm)
- aramaic-palm (arc-palm)

armenian[ul] (hy) ※
assamese[u] (as) ※
asturian[ul] (ast) ※
asu (asa) ※
atsam (cch) ※
avestan (ae) ※
awadhi (awa) ※
aymara (ay) ※
azerbaijani[ul] (az) ※
- azerbaijani-cyrillic (az-Cyrl)
- azerbaijani-cyrl (az-Cyrl)
- azerbaijani-latin (az-Latn)
- azerbaijani-latn (az-Latn)

bafia (ksf) ※
balinese (ban) ※
baluchi (bal) ※
bambara (bm) ※
bangla[u] (bn) ※
- bengali[u] (bn)

basaa (bas) ※
bashkir (ba) ※
basque[ul] (eu) ※
bataktoba (bbc) ※
bavarian (bar) ※
belarusian[ul] (be) ※
- belarusian-taraskievica[ul] (be-tarask)

bemba (bem) ※
bena (bez) ※
betawi (bew) ※
bhojpuri (bho) ※
blin (byn) ※
bodo (brx) ※
bosnian[ul] (bs) ※
- bosnian-cyrillic (bs-Cyrl)
- bosnian-cyrl (bs-Cyrl)
- bosnian-latin[ul] (bs-Latn)
- bosnian-latn[ul] (bs-Latn)

breton[ul] (br) ※
bulgarian[ul] (bg) ※
buriat[ul] (bua) ※
burmese (my) ※
cantonese (yue) ※
carian (xcr) ※
catalan[ul] (ca) ※
cebuano (ceb) ※
centralatlastamazight (tzm) ※
centralkurdish[u] (ckb) ※
- sorani[u] (ckb)
- centralkurdish-latin[u] (ckb-Latn)
- centralkurdish-latn[u] (ckb-Latn)

chakma (ccp) ※
chechen (ce) ※
cherokee (chr) ※
chiga (cgg) ※
chinese[u] (zh) ※
- chinese-simplified[u] (zh-Hans)
- chinese-hans[u] (zh-Hans)
- chinese-traditional[u] (zh-Hant)
- chinese-hant[u] (zh-Hant)
- chinese-simplified-hongkongsarchina (zh-Hans-HK)
- chinese-hans-hk (zh-Hans-HK)
- chinese-simplified-macausarchina (zh-Hans-MO)
- chinese-hans-mo (zh-Hans-MO)
- chinese-simplified-singapore (zh-Hans-SG)
- chinese-hans-sg (zh-Hans-SG)
- chinese-hant-hk (zh-Hant-HK)
- chinese-traditional-hongkongsarchina (zh-Hant-HK)
- chinese-hant-mo (zh-Hant-MO)
- chinese-traditional-macausarchina (zh-Hant-MO)
- chinese-pinyin (zh-Latn-pinyin)

churchslavic[u] (cu) ※
- churchslavic-cyrs[u] (cu-Cyrs)
- churchslavic-glag (cu-Glag)
- churchslavic-glagolitic (cu-Glag)
- churchslavic-oldcyrillic[u] (cu-Cyrs)

chuvash (cv) ※
classicalmandaic (myz) ※
colognian (ksh) ※
coptic (cop) ※
cornish (kw) ※
corsican (co) ※
croatian[ul] (hr) ※
czech[ul] (cs) ※
danish[ul] (da) ※
divehi (dv) ※
dogri (doi) ※
duala (dua) ※
dutch[ul] (nl) ※
dzongkha (dz) ※
egyptianarabic (arz) ※
: Masri or Colloquial Egyptian, with tag arz, different from Standard Arabic as spoken in Egypt, with tag ar-EG.

embu (ebu) ※
english[ul] (en) ※
- american[ul] (en-US)
- americanenglish[ul] (en-US)
- usenglish[ul] (en-US)

australian[ul] (en-AU)
australianenglish[ul] (en-AU)
british[ul] (en-GB)
britishenglish[ul] (en-GB)
ukenglish[ul] (en-GB)
canadian[ul] (en-CA)
canadianenglish[ul] (en-CA)
english-australia[ul] (en-AU)
english-au[ul] (en-AU)
english-canada[ul] (en-CA)
english-ca[ul] (en-CA)
english-unitedkingdom[ul] (en-GB)
english-gb[ul] (en-GB)
english-newzealand[ul] (en-NZ)
newzealand[ul] (en-NZ)
english-unitedstates[ul] (en-US)
english-nz[ul] (en-NZ)
english-us[ul] (en-US)
erzya (myv) ※
esperanto[ul] (eo) ※
estonian[ul] (et) ※
etruscan (ett) ※
ewe (ee) ※
ewondo (ewo) ※
faroese (fo) ※
filipino (fil) ※
finnish[ul] (fi) ※
french[ul] (fr) ※
acadian[ul] (fr-x-acadian)
canadianfrench[ul] (fr-CA)
swissfrench[ul] (fr-CH)
french-belgium[ul] (fr-BE)
french-be[ul] (fr-BE)
french-canada[ul] (fr-CA)
french-ca[ul] (fr-CA)
french-luxembourg[ul] (fr-LU)
french-lu[ul] (fr-LU)
french-switzerland[ul] (fr-CH)
french-ch[ul] (fr-CH)
friulian[ul] (fur) ※
fulah (ff) ※
ga (gaa) ※
galician[ul] (gl) ※
ganda (lg) ※
geez (gez) ※
georgian[u] (ka) ※
german[ul] (de) ※
Note the ldf names differ. See note above and also 25.17 ※ .
german-traditional[ul] (de-1901)
austrian[ul] (de-AT)
german-austria[ul] (de-AT)
german-at[ul] (de-AT)
german-austria-traditional[ul] (de-AT-1901)
swisshighgerman[ul] (de-CH)
swissgerman, with tag gsw is a different language. But see 25.17 ※ .
german-switzerland[ul] (de-CH)
german-ch[ul] (de-CH)
german-switzerland-traditional[ul] (de-CH-1901)
gothic (got) ※
greek[ul] (el) ※
monotonicgreek[ul] (el)
polytonicgreek[ul] (el-polyton)
guarani (gn) ※
gujarati[u] (gu) ※
gusii (guz) ※
haryanvi (bgc) ※
hausa[ul] (ha) ※
hausa-ghana (ha-GH)
hausa-gh (ha-GH)
hausa-niger (ha-NE)
hausa-ne (ha-NE)
hawaiian (haw) ※
hebrew[ul] (he) ※
hindi[u] (hi) ※
hmongnjua (hnj) ※
hungarian[ul] (hu) ※
magyar[ul] (hu)
icelandic[ul] (is) ※
igbo (ig) ※
inarisami (smn) ※
indonesian[ul] (id) ※
ingush (inh) ※
interlingua[ul] (ia) ※
interslavic[ul] (isv) ※
inuktitut (iu) ※
irish[ul] (ga) ※
italian[ul] (it) ※
japanese[u] (ja) ※
javanese (jv) ※
jju (kaj) ※
jolafonyi (dyo) ※
kabuverdianu (kea) ※
kabyle (kab) ※
kaingang (kgp) ※
kako (kkj) ※
kalaallisut (kl) ※
kalenjin (kln) ※
kamba (kam) ※
kangri (xnr) ※
kannada[u] (kn) ※
kashmiri (ks) ※
kazakh (kk) ※
khmer[u] (km) ※
kikuyu (ki) ※
kinyarwanda (rw) ※
komi (kv) ※
konkani (kok) ※
korean[u] (ko) ※
korean-han[u] (ko-Hani)
korean-hani[u] (ko-Hani)
koyraborosenni (ses) ※
koyrachiini (khq) ※
kurdish[ul] (ku) ※
kurdish-arab[u] (ku-Arab)
kurdish-arabic[u] (ku-Arab)
northernkurdish[ul] (ku)
northernkurdish-arab[u] (ku-Arab)
northernkurdish-arabic[u] (ku-Arab)
kwasio (nmg) ※
kyrgyz (ky) ※
ladino (lad) ※
lakota (lkt) ※
langi (lag) ※
lao[u] (lo) ※

latin[ul] (la) ※
  ecclesiasticallatin[ul] (la-x-ecclesia)
  classicallatin[ul] (la-x-classic)
  medievallatin[ul] (la-x-medieval)
latvian[ul] (lv) ※
lepcha (lep) ※
ligurian (lij) ※
limbu (lif) ※
  limbu-limb (lif-limb)
  limbu-limbu (lif-limb)
lineara (lab) ※
lingala (ln) ※
lithuanian[ulll] (lt) ※
lombard (lmo) ※
lowersorbian[ul] (dsb) ※
lowgerman (nds) ※
lu (khb) ※
lycian (xlc) ※
lydian (xld) ※
lubakatanga (lu) ※
luo (luo) ※
luxembourgish[ul] (lb) ※
luyia (luy) ※
macedonian[ul] (mk) ※
machame (jmc) ※
maithili (mai) ※
makasar (mak) ※
  makasar-bugi (mak-Bugi)
  makasar-buginese (mak-Bugi)
makhuwa (vmw) ※
makhuwameetto (mgh) ※
makonde (kde) ※
malagasy (mg) ※
malay[ul] (ms) ※
  malay-brunei (ms-BN)
  malay-bn (ms-BN)
  malay-singapore (ms-SG)
  malay-sg (ms-SG)
malayalam[u] (ml) ※
maltese (mt) ※
manipuri (mni) ※
manx (gv) ※
maori (mi) ※
marathi[u] (mr) ※
masai (mas) ※
mazanderani (mzn) ※
meru (mer) ※
meta (mgo) ※
mongolian (mn) ※
monotonicgreek[ul] (el) ※
morisyen (mfe) ※
mundang (mua) ※
muscogee (mus) ※
nama (naq) ※
navajo (nv) ※
nepali (ne) ※
newari (new) ※
ngiemboon (nnh) ※
ngomba (jgo) ※
nheengatu (yrl) ※
nigerianpidgin (pcm) ※
nko (nqo) ※
northernfrisian (frr) ※
northernluri (lrc) ※
northernsami[ul] (se) ※
  samin[ul] (se)
northernsotho (nso) ※
northndebele (nd) ※
norwegian[ul] (no) ※
  norsk[ul] (no)
    In the CLDR, norwegianbokmal (nb) just inherites from norwegian, so use the latter.
nuer (nus) ※
nyanja (ny) ※
nyankole (nyn) ※
nynorsk[ul] (nn) ※
  norwegiannynorsk[ul] (nn)
occitan[ul] (oc) ※
odia[u] (or) ※
    Preferred to oriya.
oldirish (sga) ※
oldnorse (non) ※
oldpersian (peo) ※
olduighur (oui) ※
oromo (om) ※
osage (osa) ※
ossetic (os) ※
papiamento (pap) ※
pashto (ps) ※
persian[u] (fa) ※
  farsi[u] (fa)
phoenician (phn) ※
piedmontese[ul] (pms) ※
polish[ul] (pl) ※
portuguese[ul] (pt) ※
  brazilian[ul] (pt-BR)
  brazilianportuguese[ul] (pt-BR)
  portuguese-brazil[ul] (pt-BR)
  portuguese-br[ul] (pt-BR)
  europeanportuguese[ul] (pt-PT)
  portuguese-portugal[ul] (pt-PT)
  portuguese-pt[ul] (pt-PT)
prussian (prg) ※
punjabi[u] (pa) ※
  punjabi-arabic (pa-Arab)
  punjabi-arab (pa-Arab)
  punjabi-gurmukhi[u] (pa-Guru)
  punjabi-guru[u] (pa-Guru)
quechua (qu) ※
rajasthani (raj) ※
romanian[ul] (ro) ※
  moldavian[ul] (ro-MD)
  romanian-moldova[ul] (ro-MD)
  romanian-md[ul] (ro-MD)
romansh[ul] (rm) ※
rombo (rof) ※
rundi (rn) ※
russian[ul] (ru) ※
rwa (rwk) ※
sabaean (xsa) ※
saho (ssy) ※
sakha (sah) ※
samaritan (smp) ※
samburu (saq) ※
sango (sg) ※
sangu (sbp) ※

sanskrit (sa) ※
  sanskrit-bangla (sa-Beng)
  sanskrit-beng (sa-Beng)
  sanskrit-devanagari (sa-Deva)
  sanskrit-deva (sa-Deva)
  sanskrit-gujarati (sa-Gujr)
  sanskrit-gujr (sa-Gujr)
  sanskrit-kannada (sa-Knda)
  sanskrit-knda (sa-Knda)
  sanskrit-malayalam (sa-Mlym)
  sanskrit-mlym (sa-Mlym)
  sanskrit-telugu (sa-Telu)
  sanskrit-telu (sa-Telu)
santali (sat) ※
saraiki (skr) ※
sardinian (sc) ※
scottishgaelic[ul] (gd) ※
  scottish[ul] (gd)
sena (seh) ※
serbian[ul] (sr) ※
  Note the ldf names differ. See note above.
  serbian-cyrillic[ul] (sr-Cyrl)
  serbian-cyrl[ul] (sr-Cyrl)
  serbian-cyrillic-bosniaherzegovina[ul] (sr-Cyrl-BA)
  serbian-cyrl-ba[ul] (sr-Cyrl-BA)
  serbian-cyrillic-kosovo[ul] (sr-Cyrl-XK)
  serbian-cyrl-xk[ul] (sr-Cyrl-XK)
  serbian-cyrillic-montenegro[ul] (sr-Cyrl-ME)
  serbian-cyrl-me[ul] (sr-Cyrl-ME)
  serbian-latin[ul] (sr-Latn)
  serbian-latn[ul] (sr-Latn)
  serbian-latin-bosniaherzegovina[ul] (sr-Latn-BA)
  serbian-latn-ba[ul] (sr-Latn-BA)
  serbian-latin-kosovo[ul] (sr-Latn-XK)
  serbian-latn-xk[ul] (sr-Latn-XK)
  serbian-latin-montenegro[ul] (sr-Latn-ME)
  serbian-latn-me[ul] (sr-Latn-ME)
  serbian-ijekavsk[ul] (sr-ijekavsk)
  serbian-latn-ijekavsk[ul] (sr-Latn-ijekavsk)
shambala (ksb) ※
shona (sn) ※
sichuanyi (ii) ※
sicilian (scn) ※
silesian (szl) ※
sindhi (sd) ※
  sindhi-devanagari (sd-deva)
  sindhi-deva (sd-deva)
  sindhi-khojki (sd-khoj)
  sindhi-khoj (sd-khoj)
  sindhi-khudawadi (sd-sind)
  sindhi-sind (sd-sind)
sinhala[u] (si) ※
sinteromani (rmo) ※
slovak[ul] (sk) ※
slovenian[ul] (sl) ※
  slovene[ul] (sl)
soga (xog) ※
somali (so) ※
southernaltai (alt) ※
southernsotho (st) ※
southndebele (nr) ※
spanish[ul] (es) ※
  mexican[ul] (es-MX)
  mexicanspanish[ul] (es-MX)
  spanish-mexico[ul] (es-MX)
  spanish-mx[ul] (es-MX)
standardmoroccantamazight (zgh) ※
sundanese (su) ※
swahili (sw) ※
swati (ss) ※
swedish[ul] (sv) ※
swissgerman (gsw) ※
  Different from swisshighgerman (de-CH), which is German as spoken in Switzerland. Currently still assigned to the de-CH-1901; see 25.17 ※ .
  alemannic (gsw)
  alsatian (gsw-FR)
syriac (syr) ※
tachelhit (shi) ※
  tachelhit-latin (shi-Latn)
  tachelhit-latn (shi-Latn)
  tachelhit-tifinagh (shi-Tfng)
  tachelhit-tfng (shi-Tfng)
tainua (tdd) ※
taita (dav) ※
tajik (tg) ※
tamil[u] (ta) ※
tangut (txg) ※
taroko (trv) ※
tasawaq (twq) ※
tatar (tt) ※
telugu[u] (te) ※
teso (teo) ※
thai[ul] (th) ※
tibetan[u] (bo) ※
tigre (tig) ※
tigrinya (ti) ※
tokpisin (tpi) ※
tongan (to) ※
tsonga (ts) ※
tswana (tn) ※
turkish[ul] (tr) ※
turkmen[ul] (tk) ※
tyap (kcg) ※
ugaritic (uga) ※
ukrainian[ul] (uk) ※
uppersorbian[ul] (hsb) ※
urdu[u] (ur) ※
uyghur[u] (ug) ※
uzbek (uz) ※
  uzbek-arabic (uz-Arab)
  uzbek-arab (uz-Arab)
  uzbek-cyrillic (uz-Cyrl)
  uzbek-cyrl (uz-Cyrl)
  uzbek-latin (uz-Latn)
  uzbek-latn (uz-Latn)
vai (vai) ※
  vai-latin (vai-Latn)
  vai-latn (vai-Latn)
  vai-vai (vai-Vaii)
  vai-vaii (vai-Vaii)
venda (ve) ※
venetian (vec) ※
vietnamese[ul] (vi) ※

volapuk (vo) ※
vunjo (vun) ※
walser (wae) ※
waray (war) ※
welsh[ul] (cy) ※
westernfrisian (fy) ※
wolaytta (wal) ※
wolof (wo) ※
xhosa (xh) ※
yangben (yav) ※
yiddish (yi) ※
yoruba (yo) ※
zarma (dje) ※
zhuang (za) ※
zulu (zu) ※

1.8 Fonts in Unicode engines

NOTE For pdflatex and the standard fontenc mechanism, refer to the LaTeX manuals or *The LaTeX Companion*, vol. I, ch. 9, and vol. II, ch. 10.

3.15 Babel has native support for Unicode fonts (OpenType and TrueType) in luatex and xetex by means of a high level interface on top of `fontspec`. This makes it easier to handle a wide range of languages and scripts, and simplifies the process of typesetting multilingual documents. As described below, with luatex the font can be switched automatically based on the script without explicit markup.

There is no need to load fontspec explicitly – babel does it for you with the first `\babelfont`.

NOTE See also the luaotfload fallback mechanism for a complementary approach, which can be preferable when a font lacks a character or glyph required by a language.

`\babelfont[⟨language-list⟩]{⟨font-family⟩}[⟨font-options⟩]{⟨font-name⟩}`
`\babelfont[⟨language-list⟩]{⟨font-family⟩}{⟨font-name⟩}[⟨font-options⟩]`

The main purpose of `\babelfont` is to define at once the fonts required by the different languages, with their corresponding language systems (script and language). So, if you load, say, 4 languages, `\babelfont{rm}{FreeSerif}` defines 4 fonts (with their variants, of course), which are switched with the language by babel. It is a tool to make things easier and transparent to the user. 25.15 ※ The `[⟨font-options⟩]` can go before or after the `{⟨font-name⟩}`.

Here *font-family* is `rm`, `sf` or `tt` (or newly defined ones, as explained below), and *font-name* is the same as in fontspec and the like.

If no language is given, then it is considered the default font for the family. On the other hand, if there is one or more languages in the optional argument, the font will be assigned to all of them, overriding the default one. Alternatively, you may set a font for all languages using a script – just precede the script name (lowercase) with a star (e.g., `*devanagari`).

With the optional argument, the font is *not* yet loaded, but just predeclared. In other words, font loading is lazy, which means you may define as many fonts as you want 'just in case', because if the language is never selected, the corresponding `\babelfont` declaration is just ignored.

Babel takes care of the font language and the font script when languages are selected (as well as the writing direction). In other words, there is usually no need to set the `Language` and the `Script` explicitly; in fact, it's even discouraged. In most cases, you will not need *font-options*, which is the same as in fontspec, but you may add further key/value pairs if necessary.

`\babelfont` can be used to implicitly define a new font family. Just write its name instead of `rm`, `sf` or `tt`. This is the preferred way to select fonts in addition to the three basic families.

24.14 ※ With luatex, Babel selects the renderer in the following way:

- The default renderer in alphabetic scripts, so that the tools provided by luaotfload for ligatures, kerning and the like, which are essential typographical features in these scripts, are directly available: Armenian, Coptic, Cyrillic, Georgian, Glagolitic, Gothic, Greek, Latin, Old Church Slavonic Cyrillic.

- Harfbuzz in the rest of scripts, particularly complex scripts with heavy contextual analysis (like Arabic and Devanagari). Note you can still modify fonts with 'constructors', as explained in *Modifying fonts with Harfbuzz*.

You can still select a different renderer with the fontspec key `Renderer`. Note also the same font can be loaded with different renderers. See *Comparing the modes* for further info.

NOTE There is a list of fonts in *Which method for which language*. If you know the codepoint of a character in the script you need, you can find fonts containing it with albatross (requires Java) or with something like `fc-list :charset=1033C family` in the commands line (in this case, a Gothic character, the script required by the `gothic` language).

NOTE As babel sets the font language system, the following setup is redundant, so avoid it and use only the first line:

```
\babelfont{rm}{DejaVu Serif}
\babelfont[armenian]{rm}[Script=Armenian, Language=Armenian]{DejaVu Serif}
```

EXAMPLE Usage in most cases is very simple. Let us assume you are setting up a document in Swedish, with some words in Hebrew, with a font suited for both languages. Here the option `bidi=default` activates a simple (or, rather, simplistic) bidirectional mode.

```
\documentclass{article}

\usepackage[swedish, bidi=default]{babel}

\babelfont{rm}{NewComputerModern10}

\begin{document}

Svenska \foreignlanguage{hebrew}{עִבְרִית} svenska.

\end{document}
```

If on the other hand you have to resort to different fonts, you can replace the red line above with, say:

```
\babelfont{rm}{Iwona}
\babelfont[hebrew]{rm}{FreeSerif}
```

EXAMPLE Thanks to this high level interface to `fontspec`, the roman fonts for all secondary languages in the Cyrillic and Greek scripts can be set at once with the following single line:

```
\babelfont[*cyrillic, *greek]{rm}{NewComputerModern10}
```

And if you need, say, Arabic and Devanagari:

```
  \babelfont[*arabic, *devanagari]{rm}{FreeSerif}
```

Babel does the rest for you, including setting the font script *and* language.

EXAMPLE Since each locale is a separate 'language', they can be assigned different fonts. In this example, we set Simplified and Tradicional Chinese:

```
\babelfont[chinese-simplified]{rm}{Noto Serif CJK SC}
\babelfont[chinese-simplified]{sf}{Noto Sans CJK SC}
\babelfont[chinese-traditional]{rm}{Noto Serif CJK TC}
\babelfont[chinese-traditional]{sf}{Noto Sans CJK TC}
```

EXAMPLE Here is how to declare a new family:

```
\babelfont{kai}{FandolKai}
```

Now, `\kaifamily` and `\kaidefault`, as well as `\textkai` are at your disposal.

NOTE You may load fontspec explicitly. For example:

```
\usepackage{fontspec}
\newfontscript{Devanagari}{deva}
\babelfont[hindi]{rm}{Shobhika}
```

This makes sure the OpenType script for Devanagari is `deva` and not `dev2`, in case you need it.

NOTE `\fontspec` is not touched at all, only the preset font families (`rm`, `sf`, `tt`, and the like). If a language is switched when an *ad hoc* font is active, or you select the font with this command, neither the script nor the language is passed. You must add them by hand. This is by design, for several reasons —for example, each font has its own set of features and a generic setting for several of them can be problematic, and also preserving a "lower-level" font selection is useful.

NOTE Directionality is a property affecting margins, indentation, column order, etc., not just text. Therefore, it is under the direct control of the language, which applies both the script and the direction to the text. As a consequence, there is no need to set `Script` when declaring a font with `\babelfont` (nor `Language`). In fact, it is even discouraged.

NOTE The keys `Language` and `Script` just pass these values to the *font*, and do *not* set the script for the *language* (and therefore the writing direction). In other words, the `ini` file or `\babelprovide` provides default values for `\babelfont` if omitted, but the opposite is not true. See the note above for the reasons of this behavior.

NOTE `\babelfont` is a high level interface to fontspec, and therefore in xetex you can apply `Mappings`. For example, there is a set of transliterations for Brahmic scripts by David M. Jones. After installing them in you distribution, just set the map as you would do with fontspec.

WARNING Using `\set`*xxxx*`font` and `\babelfont` at the same time is discouraged, but very often works as expected. However, be aware with `\set`*xxxx*`font` the language system will not be set by babel and should be set with `fontspec` if necessary.

TROUBLESHOOTING *Package fontspec Info: Language '<lang>' not explicitly supported within font '<font>' with script '<script>'.*

This is *not* and error. This info is shown by fontspec, not by babel. If everything is okay in your document (and almost always it is), the best thing you can do is just to ignore it altogether.

In some forums you can find the advice to set, more or less mechanically, `Language=Default`. *Do not follow it*, because font features for the language will not be applied, which can be relevant for many languages, like Urdu, Marathi and Turkish. Set the `Language` explicitly only if there is a reason to do it (for example, if fontspec/luaotfload doesn't fall back correctly a font feature to the default language). If you really want to conceal this message, use instead:

```
\PassOptionsToPackage{silent}{fontspec}
```

TROUBLESHOOTING *Package babel Info: The following fonts are not babel standard families.*

This is *not* an error. babel assumes that if you are using `\babelfont` for a family, very likely you want to define the rest of them. If you don't, you can find some inconsistencies between families. This checking is done at the beginning of the document, at a point where we cannot know which families will be used.

Actually, there is no real need to use \babelfont in a monolingual document, if you set the language system in \setmainfont (or not, depending on what you want).

As the message explains, *there is nothing intrinsically wrong* with not defining all the families. In fact, there is nothing intrinsically wrong with not using \babelfont at all. But you must be aware that this may lead to some problems.

1.9 Basic language selectors

This section describes the commands to be used in the document to switch the language in multilingual documents. In most cases, only the two basic macros \selectlanguage and \foreignlanguage are necessary. The environments otherlanguage, otherlanguage* are auxiliary, and described in the next section.

The main language is selected automatically when the document environment begins.

\selectlanguage[⟨*option-list*⟩]{⟨*language*⟩}
\begin{selectlanguage}[⟨*option-list*⟩]{⟨*language*⟩} ... \end{selectlanguage}

When a user wants to switch from one language to another he can do so using the macro \selectlanguage. It is meant for blocks of texts, and therefore should be used mainly in vertical mode, although it also works in horizontal mode.

```
\selectlanguage{german}
```

This command can be used as environment, too, in case there are relatively short texts and you do not want to reset the language with a hardcoded value.

The optional argument currently supports only nofiles. See the warning below.

NOTE Bear in mind \selectlanguage can be automatically executed, in some cases, in the auxiliary files, at heads and foots, and after the environment otherlanguage*.

WARNING If used inside braces or a group there might be some non-local changes, as it would be roughly equivalent to:

```
\bgroup
\selectlanguage{<inner-language>}
...
\egroup
\selectlanguage{<outer-language>}
```

If you want a change which is really local, you must enclose this code with an additional grouping level. The same applies to the environment version.

WARNING There are a couple of issues related to the way the language information is written to the auxiliary files:

- \selectlanguage should not be used inside some boxed environments (like floats or minipage) to switch the language if you need the information written to the aux be correctly synchronized. This rarely happens, but if it were the case, you must use the environment version or otherlanguage instead.
- In addition, this macro inserts a \write in vertical mode, which may break the vertical spacing in some cases (for example, between lists or at the beginning of a table cell). 3.64 ※ The behavior can be adjusted with \babeladjust{select.write=⟨*mode*⟩}, where ⟨*mode*⟩ is:
  - keep, the default – with it the \write and the skips are kept in the order they are written);
  - shift, which shifts the skips down and adds a \penalty;
  - omit, which may seem a too drastic solution, because nothing is written, but more often than not this command is applied to more or less shorts texts with no sectioning or similar commands, and therefore no language synchronization is necessary.

In a table cell, a `\leavevmode` just before the selector may be enough.

26.1 ※ You can also bypass file writing on a case-by-case basis using the `nofiles` optional argument, as follows:

```
\selectlanguage[nofiles]{<language>}
```

Think of it as a kind of temporary `\nofiles`, just for this command.

`\foreignlanguage[`⟨*option-list*⟩`]{`⟨*language*⟩`}{`⟨*text*⟩`}`

It takes two mandatory arguments; the second argument is a phrase to be typeset according to the rules of the language named in its first one.

This command (1) only switches the extra definitions and the hyphenation rules for the language, *not* the names and dates, (2) does not send information about the language to auxiliary files (i.e., the surrounding language is still in force), and (3) it works even if the language has not been set as package option (but in such a case it only sets the hyphenation patterns and a warning is shown). With the `bidi` option and with `\DocumentMetadata` it also enters in horizontal mode (this is not done always for backwards compatibility), and since it is meant for phrases only the text direction (and not the paragraph one) is set.

3.44 ※ As already said, captions and dates are not switched. However, with the optional argument you can switch them, too. So, you can write:

```
\foreignlanguage[date]{polish}{\today}
```

In addition, captions can be switched with `captions` (or both, of course, with `date,captions`). Until 3.43 you had to write something like `{\selectlanguage{..} ..}`, which was not always the most convenient way. See also the variant `\otherlanguage*` below.

NOTE `\bibitem` is out of sync with `\selectlanguage` in the aux file. The reason is `\bibitem` uses `\immediate` (and others, in fact), while `\selectlanguage` doesn't. There is a similar issue with floats, too. There is no known workaround, but it's not usually a real issue.

1.10 Auxiliary language selectors

`\begin{otherlanguage}{`⟨*language*⟩`}` ... `\end{otherlanguage}`

Same as `selectlanguage` as environment, except spaces after the `\begin` and `\end` commands are ignored. (Very likely, and because of the limitations of many old editors with bidi text, the idea was `\end{otherlanguage}` had to be a line by itself.) The warning above about the internal `\write` also applies here. The current mode (vertical or horizontal) also remains unchanged.

`\begin{otherlanguage*}[`⟨*option-list*⟩`]{`⟨*language*⟩`}` ... `\end{otherlanguage*}`

Same as `\foreignlanguage` but as environment. Spaces are *not* ignored. 26.1 ※ It can be used as a command, too, like `{\otherlanguage*{tamil}{...}`, but avoid using several of them inside a group.

It was originally devised for intermixing left-to-right typesetting with right-to-left typesetting in engines not supporting a change in the writing direction inside a line. However, by default it never complied with the documented behavior and it is just a version as environment of `\foreignlanguage`, except when the package option `bidi` is set or with `\DocumentMetadata` – in these cases, `\foreignlanguage` emits a `\leavevmode`, while `otherlanguage*` does not. Like `\foreignlanguage` it is a *text* command, and therefore it doesn't change the paragraph direction.

EXAMPLE Here is a document putting in practice some of the techniques described, which shows how to deal neatly with complex multilingual documents in luatex and xetex, with the help of logical markup. You are writing a book on Indic literature with many

extracts in several languages, which fits in the category of ‘mostly monolingual’. Loading of locales and fonts is lazy, which greatly simplifies the preamble.

```
\documentclass{article}

\usepackage[english]{babel}

\babelfont[*devanagari]{rm}{FreeSans}

\newenvironment{excerpt}[1]
  {\begin{quote}\begin{otherlanguage*}{#1}}
  {\end{otherlanguage*}\end{quote}}

\begin{document}

\section{Sanskrit literature}

Here is an excerpt:
\begin{excerpt}{sanskrit}
  संस्कृतम्
\end{excerpt}

\section{Hindi literature}

Here is an excerpt:
\begin{excerpt}{hindi}
  हिन्दी
\end{excerpt}

\section{Marathi literature}

Here is an excerpt:
\begin{excerpt}{marathi}
  मराठी
\end{excerpt}

\section{Nepali literature}

Here is an excerpt:
\begin{excerpt}{nepali}
  नेपाली
\end{excerpt}

\section{Rajasthani literature}

Here is an excerpt:
\begin{excerpt}{rajasthani}
  संस्कृतम्
\end{excerpt}

\end{document}
```

1.11 Plain

In e-Plain and pdf-Plain, load languages styles with `\input` and then use `\begindocument` (the latter is defined by babel):

```
\input estonian.sty
\begindocument
```

WARNING Not all languages provide a `sty` file and some of them are not compatible with those formats. Please, refer to *Using babel with Plain* for further details.

2 More on language loading and selection

2.1 A few tools

`\babeltags{`⟨*tag1*⟩ `=` ⟨*language1*⟩, ⟨*tag2*⟩ `=` ⟨*language2*⟩, …`}`

3.9i In multilingual documents with many language-switches the commands above can be cumbersome. With this tool shorter names can be defined. It adds nothing really new – it is just syntactical sugar.

It defines `\text`⟨*tag1*⟩`{`⟨*text*⟩`}` to be `\foreignlanguage{`⟨*language1*⟩`}{`⟨*text*⟩`}`, and `\begin{`⟨*tag1*⟩`}` to be `\begin{otherlanguage*}{`⟨*language1*⟩`}`, and so on. Note `\`⟨*tag1*⟩ is also allowed, but remember to set it locally inside a group.

WARNING There is a clear drawback to this feature, namely, the 'prefix' `\text...` is heavily overloaded in LaTeX and conflicts with existing macros may arise (`\textlatin`, `\textbar`, `\textit`, `\textcolor` and many others). The same applies to environments, because `arabic` conflicts with `\arabic`. Furthermore, and because of this overloading, detecting the language of a chunk of text by external tools can become unfeasible (is `\textga` the locale for the African language Gã or something else?). Except if there is a reason for this 'syntactical sugar', the best option is to stick to the default selectors or even to define your own alternatives.

EXAMPLE With

```
\babeltags{de = german}
```

you can write

```
text \textde{German text} text
```

and

```
text
\begin{de}
  German text
\end{de}
text
```

NOTE Something like `\babeltags{finnish = finnish}` is legitimate – it defines `\textfinnish` and `\finnish` (and, of course, `\begin{finnish}`).

`\babelensure[include=`⟨*commands*⟩`,exclude=`⟨*commands*⟩`,fontenc=`⟨*encoding*⟩`]{`⟨*language*⟩`}`

3.9i Except in a few languages, like russian, captions and dates are just strings, and do not switch the language. That means you should set it explicitly if you want to use them, or hyphenation (and in some cases the text itself) will be wrong. For example:

```
\foreignlanguage{russian}{text \foreignlanguage{polish}{\seename} text}
```

Of course, TeX can do it for you. To avoid switching the language all the while, `\babelensure` redefines the captions for a given language to wrap them with a selector:

```
\babelensure{polish}
```

By default only the basic captions and `\today` are redefined, but you can add further macros with the key `include` in the optional argument (without commas). Macros not to be modified are listed in `exclude`. You can also enforce a font encoding with the option `fontenc` (with it, encoded strings may not work as expected). A couple of examples:

```
\babelensure[include=\Today]{spanish}
\babelensure[fontenc=T5]{vietnamese}
```

They are activated when the language is selected (at the `afterextras` event), and it makes some assumptions which could not be fulfilled in some languages. Note also you should include only macros defined by the language, not global macros (e.g., `\TeX` of `\dag`).

With `ini` files (see below), captions are ensured by default.

Should work with Plain, but you may need to reselect the language.

`\AfterBabelLanguage{`⟨*option-name*⟩`}{`⟨*code*⟩`}`

Executes ⟨*code*⟩ when the file loaded by the corresponding package option is finished (at `\ldf@finish`). The setting is global. So

```
\AfterBabelLanguage{french}{...}
```

does `...` at the end of `french.ldf`. It can be used in `ldf` files, too, but in such a case the code is executed only if ⟨*option-name*⟩ is the same as `\CurrentOption` (which could not be the same as the option name as set in `\usepackage`!).

EXAMPLE Consider two languages foo and bar defining the same `\macro` with `\newcommand`. An error is raised if you attempt to load both. Here is a way to overcome this problem, with the help of the `presets` hook (which has to be set before loading babel):

```
\AddToHook{babel/presets}{%
  \AfterBabelLanguage{foo}{%
    \let\macroFoo\macro
    \let\macro\relax}}
\usepackage[foo, bar]{babel}
```

NOTE An alternative method to execute some code just after an `ldf` file is loaded is with the hook `file/`⟨*language*⟩`.ldf/after`. You can also execute some code before with `file/`⟨*language*⟩`.ldf/before`.

2.2 Accessing language info

`\localename`

`\mainlocalename`

`\languagename`

24.10 ※ The control sequence `\localename` contains the name of the current locale. This is now the recommended way to retrieve the current language. In addtion, `\mainlocalename` contains the main language.

`\languagename` is now discouraged at the user level (although still used internally in the babel code).

WARNING Due to a bug, which led to some internal inconsistencies in catcodes, `\languagename` should *not* be used to test which is the current language. Rely on `\localename` or, if you still need `\languagename` for some reason, on iflang, by Heiko Oberdiek.

`\iflanguage{`⟨*language*⟩`}{`⟨*true*⟩`}{`⟨*false*⟩`}`

Here "language" is used in the TEX sense, as a set of hyphenation patterns, and *not* as its babel name. The first argument is the name of a language.

`\localeinfo*{⟨field⟩}`

3.38※ If an `ini` file has been loaded for the current language, you may access the information stored in it. Note with the 'classical' `ldf` files the corresponding `ini` ones are also loaded, but only some basic data required for fonts, casing and a few more.

This macro is fully expandable, but raises an error if the info doesn't exist.

3.75※ Sometimes, it comes in handy to be able to use `\localeinfo` in a quite fully expandable way even if something went wrong (for example, the locale currently active is undefined). For these cases, `localeinfo*` just returns an empty string instead of raising an error.

The available fields are:

`name.english` as provided by the Unicode CLDR.

`tag.ini` is the tag of the `ini` file (the way this file is identified in its name).

`tag.bcp47` is the full BCP 47 tag. This is the value to be used for the 'real' provided tag (babel may fill other fields if they are considered necessary).

`language.tag.bcp47` is the BCP 47 language tag.

`tag.opentype` is the tag used by OpenType (usually, but not always, the same as BCP 47). It can be useful when customizing fonts with luaotfload.

`script.name` as provided by the Unicode CLDR.

`script.tag.bcp47` is the BCP 47 tag of the script used by this locale. This is a required field for the fonts to be correctly set up, and therefore it should always be defined.

`script.tag.opentype` is the tag used by OpenType (usually, but not always, the same as BCP 47). It can be useful when customizing fonts with luaotfload.

`region.tag.bcp47` is the BCP 47 tag of the region or territory. Defined only if the locale loaded actually contains it (e.g., `es-MX` does, but `es` doesn't), which is how locales behave in the CLDR. 3.75※

`variant.tag.bcp47` is the BCP 47 tag of the variant (in the BCP 47 sense, like `1901` for German). 3.75※

`extension.⟨s⟩.tag.bcp47` is the BCP 47 value of the extension whose singleton is ⟨*s*⟩ (currently the recognized singletons are `x`, `t` and `u`). The internal syntax can be somewhat complex, and this feature is still somewhat tentative. An example is classicallatin which sets `extension.x.tag.bcp47` to `classic`. 3.75※

NOTE Currently, `x` is used for two separate functions, namely, identifying a babel locale without a BCP 47 tag and setting an alternative set of rules for casing.

NOTE Bear in mind that babel, following the CLDR, may leave the region unset, which means `\getlocaleproperty*`, described below, is the preferred command, so that the existence of a field can be checked before. This also means building a string with the language and the region with something like `\localeinfo*{language.tag.bcp47}-\localeinfo*{region.tag.bcp47}` is not usually a good idea (because of the hyphen).

WARNING 3.46※ As of version 3.46 `tag.bcp47` returns the full BCP 47 tag. Formerly it returned just the language subtag, which was clearly counterintuitive.

`\getlocaleproperty*{⟨macro⟩}{⟨locale⟩}{⟨property⟩}`

3.42※ The value of any locale property as set by the `ini` files (or added/modified with `\babelprovide`) can be retrieved and stored in a macro with this command. For example, after:

```
\getlocaleproperty\hechap{hebrew}{captions/chapter}
```

the macro `\hechap` will contain the string פרק.

If the key does not exist, the macro is set to `\relax` and an error is raised. 3.47※ With the starred version no error is raised, so that you can take your own actions with undefined properties.

`\localeid`

Each language in the babel sense has its own unique numeric identifier, which can be retrieved with `\localeid`.

The `\localeid` is not the same as the `\language` identifier, which refers to a set of hyphenation patterns (which, in turn, is just a component of the line breaking algorithm described in the next section). The data about preloaded patterns are store in an internal macro named `\bbl@languages` (see the code for further details), but note several locales may share a single `\language`, so they are separated concepts. In luatex, the `\localeid` is saved in each node (when it makes sense) as an attribute, too.

`\ShowLocaleProperties{`⟨*language*⟩`}`

3.98 ※ Prints to the `log` file all the loaded key/value pairs from the `ini` locale file for ⟨*language*⟩.

`\LocaleForEach{`⟨*code*⟩`}`

Babel remembers which `ini` files have been loaded. There is a loop named `\LocaleForEach` to traverse the list, where `#1` is the name of the current item, so that `\LocaleForEach{\message{ **#1** }}` just shows the loaded `ini`'s.

2.3 Package options

3.9a These package options are processed before language options, so that they are taken into account irrespective of their order. The first three options have been available in previous versions.

`KeepShorthandsActive`

Tells babel not to deactivate shorthands after loading a language file, so that they are also available in the preamble.

`headfoot=`⟨*language*⟩

By default, headlines and footlines are not touched (only marks), and if they contain language-dependent macros (which is not usual) there may be unexpected results. With this option you may set the language in heads and feet. An alternative is to set the language explicitly when heads and feet are redefined.

`noconfigs`

Global and language default config files are not loaded, so you can make sure your document is not spoilt by an unexpected `.cfg` file. However, if the key `config` is set, this file is loaded.

`config=`⟨*file*⟩

Load ⟨*file*⟩`.cfg` instead of the default config file `bblopts.cfg` (forced even with `noconfigs`).

WARNING 25.14 ※ Configuration files are deprecated, as they can break document portability.

`showlanguages`

Prints to the log the list of languages loaded when the format was created: number (remember dialects can share it), name, hyphenation file and exceptions file.

`silent`

3.9l No warnings and no *infos* are written to the log file. (You can use alternatively the package silence.)

`hyphenmap=off | first | select | other | other*`

3.9g Sets the behavior of case mapping for hyphenation, provided the language defines it (turned off in plain). It can take the following values:

`off` deactivates this feature and no case mapping is applied;
`first` sets it at the first switching commands in the current or parent scope (typically, when the aux file is first read and at `\begin{document}`, but also the first `\selectlanguage` in the preamble), and it's the default if a single language option has been stated. If a language option is repeated (e.g., `malay, malay`), it counts as several ones, even if there is only a language.
`select` sets it only at `\selectlanguage`;
`other` also sets it at `otherlanguage`;
`other*` also sets it at `otherlanguage*` as well as in heads and feet (if the option `headfoot` is used) and in auxiliary files (i.e., at `\select@language`), and it's the default if several language options have been stated. The option `first` can be regarded as an optimized version of `other*` for monolingual documents.[1]

`bidi=default | basic | basic-r | bidi-l | bidi-r`

3.14 Selects the bidi algorithm to be used in luatex and xetex. See sec. 5.10.

`layout=`

3.16 Selects which layout elements are adapted in bidi documents. See sec. 5.10.

`provide=* | ⟨option-list⟩`

3.49 ※ An alternative to `\babelprovide` for languages passed as options. See section 2.5, which describes also the variants `provide+=` and `provide*=`.

`main=⟨language⟩`

Forces the main language. This option should *not* be used except if there a need to do it. If the language is not given as such as package or global option, it is added to the list of requested languages. For example:

```
\documentclass{article}
\usepackage[main=english,dutch]{babel}
```

A situation where this key can be useful is with some classes loading babel with a hardcoded language option. Sometimes, the main language can be overridden with something like that before `\documentclass`:

```
\PassOptionsToPackage{main=english}{babel}
```

`licr=extended | unextended`

26.7 ※ Some languages may require extending the built-in declarations in LaTeX for the TU encoding, because a package may rely solely on them (like `\cyrzh` instead of ж). This is now done with the Cyrillic locales (further scripts may be added in the future). You can deactivate these extra declarations (in case you have devised your own, for example) with `licr=unextended` (the default is `extended`). They are ignored with lazy loading, so if you need them you have to declare explicitly the language in the preamble.

[1] Providing `foreign` is pointless, because the case mapping applied is that at the end of the paragraph, but if either luatex or xetex change this behavior it might be added. On the other hand, `other` is provided even if I [JBL] think it isn't really useful, but who knows.

2.4 The base option

With this package option babel just loads some basic macros (mainly the selectors), defines `\AfterBabelLanguage` and exits. It also selects the hyphenation patterns for the last language passed as option (by its name in `language.dat`). It doesn't load any `ldf` or `ini` files (and, in fact, many supporting macros for these files remain undefined).

WARNING This option is not compatible with languages loaded on the fly or with `\DocumentMetadata`.

2.5 `provide` and `\babelprovide` – `ini` files

[To be revised.]

An alternative approach to define a language (or, more precisely, a *locale*) is by means of an `ini` file. Currently babel provides about 380 of these files containing the basic data required for a locale, covering about 300 languages and 40 scripts, plus basic templates for about 400 locales.

`ini` files are not meant only for babel, and they has been devised as a resource for other packages. To ease interoperability between TEX and other systems, they are identified with the BCP 47 codes as preferred by the Unicode Common Locale Data Repository, which was used as source for most of the data provided by these files, where available.

Most of them set the date, and many also the captions (Unicode and LICR). They will be evolving with the time to add more features (something to keep in mind if backward compatibility is important). The following section shows how to make use of them by means of `\babelprovide`.

Not all languages in the CLDR are complete, and therefore neither are the `ini` files. A few languages may show a warning about the current lack of suitability of some features.

EXAMPLE There is an example of how to use an `ini` template file here, for Phoenician (although currently this locale is bundled with babel).

NOTE The `ini` files just define and set some parameters, but the corresponding behavior is not always implemented. Also, there are some limitations in the engines. A few remarks follow (which could no longer be valid when you read this manual, if the packages involved have been updated).

Arabic Math and graphical elements like `picture` are complex and requires some additional fine tuning. In xetex babel resorts to the bidi package, which seems to work.

Southeast scripts Thai works in both luatex and xetex, but line breaking differs (rules are hard-coded in xetex, but they can be modified in luatex). Lao seems to work, too, but there are no patterns for the latter in luatex. Some quick patterns can help, with something similar to:

```
\babelprovide[import, hyphenrules=+]{lao}
\babelpatterns[lao]{1ก 1ม 1ə 1ŋ 1ท 1ร} % Random
```

East Asia scripts Settings for either Simplified of Traditional (Chinese) should work out of the box, with basic line breaking with any renderer. Although for a few words and shorts texts the `ini` files should be fine, CJK texts are best set with a dedicated framework (CJK, luatexja, kotex, CTeX, etc.). This is what the class `ltjbook` does with luatex, which can be used in conjunction with the `ldf` for `japanese`, because the following piece of code loads luatexja:

```
\documentclass[japanese]{ltjbook}
\usepackage{babel}
```

Latin, Greek, Cyrillic Combining chars with the default luatex font renderer might be wrong; on the other hand, with the Harfbuzz renderer diacritics are stacked correctly, but many hyphenations points are discarded (this bug is related to

kerning, so it depends on the font). With xetex both combining characters and hyphenation work as expected (not quite, but in most cases it works; the problem here are font clusters).

2.6 Selection based on BCP 47 tags

3.43 ※ The recommended way to select languages is that described at the beginning of this document. However, BCP 47 tags are becoming customary, particularly in documents (or parts of documents) generated by external sources, and therefore babel will provide a set of tools to select the locales in different situations, adapted to the particular needs of each case. Currently, babel provides autoloading of locales as described in this section. In these contexts autoloading is particularly important because we may not know beforehand which languages will be requested.

It must be activated explicitly, because it is primarily meant for special tasks. Mapping from BCP 47 codes to locale names is not hardcoded in babel. Instead the data is taken from the `ini` files, which means currently about 350 tags are already recognized. Babel performs a simple lookup in the following way: `fr-Latn-FR` ▸ `fr-Latn` ▸ `fr-FR` ▸ `fr`. Languages with the same resolved name are considered the same. Case is normalized before, so that `fr-latn-fr` ▸ `fr-Latn-FR`. If a tag and a name overlap, the tag takes precedence.

Here is a minimal example:

```
\documentclass{article}

\usepackage[danish]{babel}

\babeladjust{
  autoload.bcp47 = on
}

\begin{document}

Chapter in Danish: \chaptername.

\selectlanguage{de-AT}

\localedate{2020}{1}{30}

\end{document}
```

Currently the locales loaded are based on the `ini` files and decoupled from the main `ldf` files. This is by design, to ensure code generated externally produces the same result regardless of the languages requested in the document, but an option to use the `ldf` instead will be added in a future release, because both options make sense depending on the particular needs of each document (there will be some restrictions, however).

The behavior is adjusted with `\babeladjust` with the following parameters:

`autoload.bcp47` with values `on` and `off`.

`autoload.bcp47.options` , which are passed to `\babelprovide`; 24.14 ※ `import` by default (features defined in the corresponding `babel-...tex` file might not be available), but you can set it to another value (even empty).

`autoload.bcp47.prefix` . Although the public name used in selectors is the tag, the internal name will be different and generated by prepending a prefix, which by default is `bcp47-`. You may change it with this key.

3.46 ※ If an `ldf` file has been loaded, you can enable the corresponding language tags as selector names with:

```
\babeladjust{ bcp47.toname = on }
```

(You can deactivate it with `off`.) So, if `dutch` is one of the package (or class) options, you can write `\selectlanguage{nl}`. Note the language name does not change (in this

example is still dutch), but you can get it with \localeinfo or \getlocaleproperty. It must be turned on explicitly for similar reasons to those explained above.

3 Tailoring, customizing and modifying a language

Modifying the behavior of a language (say, the chapter "caption") is sometimes necessary.

Several language definition files use their own methods to set options. For example, french uses \frenchsetup, magyar (1.5) uses \magyarOptions; modifiers provided by spanish have no attribute counterparts. Macros setting options are also used (e.g., \ProsodicMarksOn in latin).

This section describes the general tools provided by the babel core.

3.1 Captions

This is perhaps the most frequent change, so a specific macro is provided.

\setlocalecaption{⟨*language-name*⟩}{⟨*caption-name*⟩}{⟨*string*⟩}

3.51 ※ Here *caption-name* is the name as string without the trailing name. An example, which also shows caption names are often a stylistic choice, is:

```
\setlocalecaption{english}{contents}{Table of Contents}
```

This works not only with existing caption names, because it also serves to define new ones by setting the *caption-name* to the name of your choice (name will be postpended). Captions so defined or redefined behave with the 'new way' described in the following note. Should work with Plain, but remember you may need to reselect the language.

Note the string should not contain a language selector or changes in the text direction, which is done by babel when necessary. With arabic, all you need is:

```
\setlocalecaption{arabic}{part}{القسم}
```

NOTE There are a few alternative methods:

- With data import'ed from ini files, you can modify the values of specific keys when loaded, like:

  ```
  \babelprovide[import, captions/listtable = Lista de tablas]{spanish}
  ```

  (In this particular case, instead of the captions group you may need to modify the captions.licr one.)

- The low-level 'old way' to redefine a caption, still valid for many languages but discouraged in general, is the following:

  ```
  \addto\captionsenglish{%
    \renewcommand\contentsname{Foo}%
  }
  ```

  As of 3.15, there is no need to hide spaces with % (babel removes them), but it is advisable to do so. This redefinition is not activated until the language is selected.

- The 'new way', which is found in bulgarian, azerbaijani, spanish, french, turkish, icelandic, vietnamese and a few more, as well as in languages created with \babelprovide and its key import, or with the package option provide=, is:

  ```
  \renewcommand\spanishchaptername{Foo}
  ```

  This redefinition is immediate. In these languages, the month names can be redefined in a similar way; names are \⟨*language*⟩month⟨*roman-num*⟩name, where

⟨*roman-num*⟩ is the month number in lowercase roman (e.g., `\spanishmonthxiiname`).

NOTE Do *not* redefine a caption in the following way:

```
\AtBeginDocument{\renewcommand\contentsname{Foo}}
```

The changes may be discarded with a language selector, and the original value restored.

Macros to be run when a language is selected can be added to `\extras`⟨*language*⟩:

```
\addto\extrasrussian{\mymacro}
```

There is a counterpart for code to be run when a language is unselected: `\noextras`⟨*language*⟩.

NOTE These macros (`\captions`⟨*language*⟩, `\extras`⟨*language*⟩) may be redefined, but *must not* be used as such – they just pass information to babel, which executes them in the proper context.

Another way to modify a language loaded as a package or class option is by means of `\babelprovide`, described below in depth. So, something like:

```
\usepackage[danish]{babel}
\babelprovide[hyphenrules=nohyphenation]{danish}
```

first loads `danish.ldf`, and then prevents hyphenation for this locale. The rest of the language definitions are not touched. Without the optional argument it just loads some additional tools if provided by the `ini` file, like extra counters.

3.2 Modifiers

3.9c The basic behavior of some languages can be modified when loading babel by means of *modifiers*. They are set after the language name, and are prefixed with a dot (only when the language is set as package option – neither global options nor the `main` key accepts them). An example is (spaces are not significant and they can be added or removed):[2]

```
\usepackage[latin.lowercasemonth, spanish.notilde.lcroman, danish]{babel}
```

Attributes (described below) are considered modifiers, i.e., you can set an attribute by including it in the list of modifiers.

3.89 ※ Alternatively, modifiers can be set with a separate option, with the keyword `modifiers` followed by a dot and the language name (note the language is not selected or loaded with this option). It is useful to activate some feature when the language is declared as a class option:

```
\documentclass[spanish]{report}
\usepackage[modifiers.spanish = notilde.lcroman]{babel}
```

3.3 Language attributes

`\languageattribute`{⟨*language*⟩}

This is a user-level command, to be used in the preamble of a document (after `\usepackage[...]{babel}`), that declares which attributes are to be used for a given language. It takes two arguments: the first is the name of the language; the second, a (list of) attribute(s) to be used. Attributes must be set in the preamble and only once – they cannot be turned on and off. The command checks whether the language is known in this document and whether the attribute(s) are known for this language.

[2] No predefined "axis" for modifiers is provided because languages and their scripts have quite different needs.

3.4 Casing

`\BabelUppercaseMapping{`⟨*locale-name*⟩`}{`⟨*codepoint*⟩`}{`⟨*output*⟩`}`
`\BabelLowercaseMapping{`⟨*locale-name*⟩`}{`⟨*codepoint*⟩`}{`⟨*output*⟩`}`
`\BabelTitlecaseMapping{`⟨*locale-name*⟩`}{`⟨*codepoint*⟩`}{`⟨*output*⟩`}`

3.90 ※ These macros are devised as a high level interface for declaring case mappings, based on the locale name as declared by or with babel. They are the equivalent of `\DeclareUppercaseMapping`, `\DeclareLowercaseMapping`, and `\DeclareTitlecaseMapping` (see usrguide.pdf). The purpose is twofold: (1) a user-friendly way to declare them, because often BCP 47 tags are not known (and sometimes can be complex); (2) if for some reason the tag changes (e.g., you decide to tag `english` as `en-001` instead of `en-US`), the new mappings will still be assigned to that language.

EXAMPLE For Classical Latin (no need to know the tag is `la-x-classic`):

```
\BabelUppercaseMapping{classicallatin}{`u}{V}
```

NOTE There are still some rough edges when declaring a mapping with the `x` extension, or when two babel languages share the same BCP 47 tag. These issues are expected be sorted out in future releases.

3.5 Modifying and adding values to `ini` files

3.39 ※ There is a way to modify or even add values of `ini` files when they are imported with `\babelprovide` or `provide=`. This can be used to create private variants easily. All you need is to import the same `ini` file with a different locale name and different parameters. Without `import` you may still modify the identification keys.

Examples of the fine-grained control you have at your disposal are the following. All of them assume `\usepackage[english]{babel}`.

EXAMPLE Here is how to change the hyphenation rules because there are several criteria, or you must follow an editorial style. The following example just uses the default Spanish rules in English:

```
\babelprovide[hyphenrules=spanish]{english}
```

EXAMPLE The required native digits are already defined in the corresponding `ini` files, but they can be modified and even added as shown:

```
\babelprovide[numbers/digits.native=abcdefghij]{english}
```

This example is somewhat absurd, but now `\englishdigits{264}` will print ‘cge’. (It doesn’t work with pdftex.)

EXAMPLE Currently the date format can be changed only with imported data, but with its high level interface it’s rather straightforward:

```
\babelprovide[
    import,
    date.gregorian/date.long = {[d] ([MMMM]) [y]}]
  {english}
```

Here `[d]` is the day number, `[MMMM]` the month name and `[y]` the year. It will print something like ‘5 (October) 2024’. Remember `[d]` and `[y]` admit the syntax [⟨*period*⟩|⟨*counter*⟩], where ⟨*counter*⟩ is a counter name as defined in the counters section. See 3.44 ※ and the Greek `ini` files for examples.

EXAMPLE To set the hyphen to ‘none’ (only luatex).

```
\babelprovide[typography/prehyphenchar = 0]{english}
```

(This setting *may* work with `xetex`, but getting rid of the hyphen char in this engine is not trivial, because you must rely on the font, and not all fonts behave the same.)

If, on the other hand, the language does not write by default the hyphens when breaking lines and you want them, you can set `typography/prehyphenchar = `- (they are Kannada, Malayalam, Odia, Sanskrit, Tamil, Telugu).

EXAMPLE You can define new counters freely, and assign them to `\alph`:

```
\babelprovide[
    counters/acute = á é í ó ú, % Define a counter named `acute`
    alph = acute                % Assign it to \alph
  ]{english}
```

You can choose the name, and instead of `acute` it can be another one. Unlike `digits.native`, the elements are separated with spaces.

3.6 Hooks

3.9a A hook is a piece of code to be executed at certain events. Some hooks are predefined when luatex and xetex are used.

3.64 ※ This is not the only way to inject code at those points. The events listed below can be used as a hook name in `\AddToHook` in the form `babel/`⟨*language-name*⟩`/`⟨*event-name*⟩ (with * as ⟨*language-name*⟩ it's applied to all languages), but there is a limitation, because the parameters passed with the babel mechanism are not allowed. The `\AddToHook` mechanism does *not* replace the current one in babel. Its main advantage is you can reconfigure babel even before loading it. See the example below.

`\AddBabelHook[`⟨*language*⟩`]{`⟨*name*⟩`}{`⟨*event*⟩`}{`⟨*code*⟩`}`

`\EnableBabelHook{`⟨*name*⟩`}`

`\DisableBabelHook{`⟨*name*⟩`}`

The same name can be applied to several events. Hooks with a certain `{`⟨*name*⟩`}` may be enabled and disabled for all defined events with `\EnableBabelHook{`⟨*name*⟩`}`, `\DisableBabelHook{`⟨*name*⟩`}`. Names containing the string `babel` are reserved (they are used, for example, by `\useshortands*` to add a hook for the event `afterextras`).

3.33 They may be also applied to a specific language with the optional argument; language-specific settings are executed after global ones.

Current events are the following; in some of them you can use one to three TEX parameters (#1, #2, #3), with the meaning given:

`adddialect` (language name, dialect name) Used by `luababel.def` to load the patterns if not preloaded.

`patterns` (language name, language with encoding) Executed just after the `\language` has been set. The second argument has the patterns name actually selected (in the form of either `lang:ENC` or `lang`).

`hyphenation` (language name, language with encoding) Executed locally just before exceptions given in `\babelhyphenation` are actually set.

`defaultcommands` Used (locally) in `\StartBabelCommands`.

`encodedcommands` (input, font encodings) Used (locally) in `\StartBabelCommands`. Both luatex and xetex make sure the encoded text is read correctly.

`stopcommands` Used to reset the above, if necessary.

`write` This event comes just after the switching commands are written to the aux file.

`beforeextras` Just before executing `\extras`⟨*language*⟩. This event and the next one should not contain language-dependent code (for that, add it to `\extras`⟨*language*⟩).

`afterextras` Just after executing `\extras`⟨*language*⟩. For example, the following deactivates shorthands in all languages:

```
\AddBabelHook{noshort}{afterextras}{\languageshorthands{none}}
```

`stringprocess` Instead of a parameter, you can manipulate the macro `\BabelString` containing the string to be defined with `\SetString`. For example, to use an expanded version of the string in the definition, write:

```
\AddBabelHook{myhook}{stringprocess}{%
  \protected@edef\BabelString{\BabelString}}
```

`initiateactive` (char as active, char as other, original char) 3.9i Executed just after a shorthand has been 'initiated'. The three parameters are the same character with different catcodes: active, other (`\string`'ed) and the original one.

`afterreset` 3.9i Executed when selecting a language just after `\originalTeX` is run and reset to its base value, before executing `\captions`⟨*language*⟩ and `\date`⟨*language*⟩.

`begindocument` 3.88 ※ Executed before the code written by `ldf` files with `\AtBeginDocument`. The optional argument with the language in this particular case is the language that wrote the code. The special value `/` means 'return to the core babel definitions' (in other words, what follows hasn't been written by any language).

`foreign` 24.8 ※ Executed by `\foreignlanguage` after the language has been set up and just before typesetting the text from the second argument. Its main purpose it to wrap the text with some code, with the help of `\BabelWrapText`. For example, with:

```
\AddBabelHook{one}{foreign}{\BabelWrapText{\textit{##1}}
\AddBabelHook{two}{foreign}{\BabelWrapText{\parse{##1}}
```

the text becomes `\textit{\parse{`⟨*text*⟩`}}`. Order is relevant.

EXAMPLE PDF tagging is still a work in progress, but language tags are already possible in short texts with `\foreignlanguage`. See the babel site for a complete explained example.

Four events are used in `hyphen.cfg`, which are handled in a quite different way for efficiency reasons – unlike the precedent ones, they only have a single hook and replace a default definition.

`everylanguage` (language) Executed before every language patterns are loaded.

`loadkernel` (file) By default just defines a few basic commands. It can be used to define different versions of them or to load a file.

`loadpatterns` (patterns file) Loads the patterns file. Used by `luababel.def`.

`loadexceptions` (exceptions file) Loads the exceptions file. Used by `luababel.def`.

EXAMPLE The generic unlocalized LaTeX hooks are predefined, so that you can write:

```
\AddToHook{babel/*/afterextras}{\frenchspacing}
```

which is executed always after the extras for the language being selected (and just before the non-localized hooks defined with `\AddBabelHook`).

In addition, locale-specific hooks in the form `babel/`⟨*language-name*⟩`/`⟨*event-name*⟩ are *recognized* (executed just before the localized babel hooks), but they are *not predefined*. You have to do it yourself. For example, to set `\frenchspacing` only in `bengali`:

```
\ActivateGenericHook{babel/bengali/afterextras}
\AddToHook{babel/bengali/afterextras}{\frenchspacing}
```

3.7 Manage auxiliary files

`\BabelContentsFiles`

3.9a This macro contains a list of "toc" types requiring a command to switch the language. Its default value is `toc,lof,lot`, but you may redefine it with `\renewcommand` to

add further types in case you need or there is a package enabling additional files, e.g., for theorems, algorithms, notation (it's up to you to make sure no toc type is duplicated).

3.8 Code based on the selector

`\IfBabelSelectorTF{⟨selectors⟩}{⟨true⟩}{⟨false⟩}`

3.67 ※ Sometimes a different setup is desired depending on the selector used. Values allowed in ⟨*selectors*⟩ are `select`, `other`, `foreign`, `other*` (and also `foreign*` for the tentative starred version), and it can consist of a comma-separated list. For example:

```
\IfBabelSelectorTF{other, other*}{A}{B}
```

is true with any of these two environment selectors.
Its natural place of use is in hooks or in `\extras`⟨*language*⟩.

3.9 Presets

24.12 ※ Apart from configuration files, which are loaded before the locale files, you can preset some options even before loading babel. One on the aims of this feature is to ease the integration with automated document generation or conversion workflows.

`\AddToHook{babel/presets}{⟨settings⟩}`

This LaTeX hook is executed just before locale files (either `ini` or `ldf`) are loaded. It's in fact, similar to the config files, but it's executed a little later and there is no need for a separate file. The following command has been devised for this hook, but other candidates are `\AfterBabelLanguage` and `\DeclareOption` (although the latter can be somewhat dangerous).

Note these declarations are valid 'just in case'. If babel is not loaded, they are ignored.

`\PassOptionsToLocale{⟨option-list⟩}{⟨locale-name⟩}`

Its purpose is what its name suggests, and it was devised to be used with the previous hook.

EXAMPLE You are writing a class and expect lazy loading of secondary languages. You also want to make sure `french`, if used, activates its rules for punctuation spacing, and `malayalam`, if used, maps digits to the native ones (with luatex):

```
\AddToHook{babel/presets}{%
  \PassOptionsToLocale{transforms=punctuation.space}{french}%
  \PassOptionsToLocale{mapdigits}{malayalam}}
```

If you want to take a step further and force babel to always use `ini` files in all secondary languages, you can resort to the LaTeX mechanism to pass options to packages:

```
  \PassOptionsToPackage{provide+=*}{babel}
```

NOTE Remember localized fonts are preset, too, with lazy loading. In the previous example you can set, for example, `\babelfont[malayalam]{rm}{FreeSerif}`.

4 Creating a language

[To be revised, as this command is also used with existing `ini` locale files.]

3.10 And what if there is no style for your language or none fits your needs? You may then define quickly a language with the help of the following macro in the preamble (which may be used to modify an existing language, too, as explained in previous section).

`\babelprovide`[⟨*options*⟩]{⟨*language-name*⟩}

If the language ⟨*language-name*⟩ has not been loaded as class or package option and there are no ⟨*options*⟩, it creates an "empty" one with some defaults in its internal structure: the hyphen rules, if not available, are set to the current ones, left and right hyphen mins are set to 2 and 3. In either case, caption, date and language system are not defined.

If no `ini` file is imported with `import`, ⟨*language-name*⟩ is still relevant because in such a case the hyphenation and like breaking rules (including those for South East Asian and CJK) are based on it as provided in the `ini` file corresponding to that name; the same applies to OpenType language and script.

Conveniently, some options allow to fill the language, and babel warns you about what to do if there is a missing string. Very likely you will find alerts like this in the `log` file:

```
Package babel Warning: \chaptername not set for 'mylang'. Please,
(babel)                define it after the language has been loaded
(babel)                (typically in the preamble) with:
(babel)                \setlocalecaption{mylang}{chapter}{..}
(babel)                Reported on input line 26.
```

In most cases, you will only need to define a few macros. Note languages loaded on the fly are not yet available in the preamble.

EXAMPLE If you need a language named `arhinish`:

```
\usepackage[danish]{babel}
\babelprovide{arhinish}
\setlocalecaption{arhinish}{chapter}{Chapitula}
\setlocalecaption{arhinish}{refname}{Refirenke}
\babelhyphenmins[arhinish]{2}{2}
```

EXAMPLE Sometimes treating the IPA as a language makes sense:

```
\documentclass{article}
\usepackage[english]{babel}
\babelprovide{ipa}
\babelfont[ipa]{rm}{DejaVu Sans}
\begin{document}
Blah \foreignlanguage{ipa}{ɔːl'ðəʊ} Blah.
\end{document}
```

Then you can define shorthands, transforms (with luatex), interchars (with xetex) and so on, specific to this new 'language'.

EXAMPLE Locales with names based on BCP 47 codes can be created with something like:

```
  \babelprovide[import=en-US]{en-US}
```

Note, however, mixing ways to identify locales can lead to problems. For example, is `ga` the name of the language spoken by the Gã people or is it the code for Irish?

The main language is not changed (`danish` in this example). So, you must add `\selectlanguage{arhinish}` or other selectors where necessary.

If the language has been loaded as an argument in `\documentclass` or `\usepackage`, then `\babelprovide` redefines the requested data.

`import=`⟨*locale-tag*⟩

The ⟨*locale-tag*⟩ is optional. It imports the full data from the `ini` file corresponding to the locale being loaded, as set in `babel-`⟨*locale*⟩`.tex` (where ⟨*locale*⟩ is the last argument in `\babelprovide`), including captions and date (and also line breaking rules in newly

defined languages). Unicode engines load the UTF-8 variants, while 8-bit engines load the LICR (i.e., with macros like `\'` or `\ss`) ones.

However, in some cases you may want to load a different `ini` file. In such a case, you can set its value.

EXAMPLE For example, the locale `ar-DZ` is named `arabic-algeria`. You may prefer the shorter name `arabic` if there are no conflicts:

```
\babelprovide[import=ar-DZ]{arabic}
```

Besides `\today`, this option defines an additional command for dates: `\`⟨*language*⟩`date`, which takes three arguments, namely, year, month and day numbers. In fact, `\today` calls `\`⟨*language*⟩`today`, which in turn calls `\`⟨*language*⟩`date{\the\year}{\the\month}{\the\day}`. 3.44 ※ More convenient is usually `\localedate`, which prints the date for the current locale.

`hyphenrules=`⟨*language-list*⟩

With this option, with a space-separated list of hyphenation rules, babel assigns to the language the first valid hyphenation rules in the list. For example:

```
\babelprovide[hyphenrules=chavacano spanish italian]{chavacano}
```

If none of the listed hyphenrules exist, the default behavior applies. Note in this example we set `chavacano` as first option, which can seem redundant, but without it, it would select `spanish` even if `chavacano` exists.

A special value is `+`, which allocates a new language (in the TEX sense). It only makes sense as the last value (or the only one; the subsequent ones are silently ignored). It is mostly useful with luatex, because you can add some patterns with `\babelpatterns`, as for example:

```
\babelprovide[hyphenrules=+]{neo}
\babelpatterns[neo]{a1 e1 i1 o1 u1}
```

In other engines it just suppresses hyphenation (because the pattern list is empty).

3.58 ※ Another special value is `unhyphenated`, which is an alternative to `justification=unhyphenated`.

`main`

This valueless option makes the language the main one (thus overriding that set when babel is loaded).

EXAMPLE Let's assume your document (luatex or xetex) is mainly in Polytonic Greek but with some sections in Italian. Then, the first attempt should be:

```
\usepackage[italian, greek.polytonic]{babel}
```

But if, say, accents in Greek are not shown correctly, you can try

```
\usepackage[italian, polytonicgreek, provide=*]{babel}
```

Remember there is an alternative syntax for the latter:

```
\usepackage[italian]{babel}
\babelprovide[import, main]{polytonicgreek}
```

Finally, also remember you might not need to load `italian` at all if there are only a few word in this language (see 1.3).

`script=`⟨*script-name*⟩

3.15 Sets the script name to be used by fontspec (e.g., `Devanagari`). Overrides the value in the `ini` file. If fontspec does not define it, then babel sets its tag to that provided by the `ini` file. This value is particularly important because it sets the writing direction, so you must use it if for some reason the default value is wrong.

`language=`⟨*language-name*⟩

3.15 Sets the language name to be used by fontspec (e.g., `Hindi`). Overrides the value in the `ini` file. If fontspec does not define it, then babel sets its tag to that provided by the `ini` file. Not so important, but sometimes still relevant.

`alph=`⟨*counter-name*⟩

Assigns to `\alph` that counter. See the next section.

`Alph=`⟨*counter-name*⟩

Same for `\Alph`.

`casing=`⟨*rules*⟩

3.90 ※ Selects the casing rules in a few languages. The first ones are predefined by LaTeX (see interface3.pdf), while the following are defined by babel:

Armenian `yiwn` maps U+0587 to capital ech and yiwn on uppercasing.

German `eszett` maps the lowercase *Eszett* to a *großes Eszett*.

Greek `iota` converts the *ypogegrammeni* (subscript muted iota) to capital iota when uppercasing.

Latin `nouv` in `classicallatin` and `medievallatin` reverts the default rules, which maps u ‣ V; `uv` in `ecclesiasticallatin` and the basic `latin` locale applies the map u ‣ V (by default it's u ‣ U and v ‣ V).

EXAMPLE For the latter:

```
\usepackage[greek]{babel}
\babelprovide[casing=iota]{greek}
```

`transforms=`⟨*transform-list*⟩

(luatex) See section 5.7.

`interchar=`⟨*interchar-list*⟩

(xetex) See section 5.8.

NOTE (1) If you need shorthands, you can define them with `\useshorthands` and `\defineshorthand` as described above. (2) Captions and `\today` are "ensured" with `\babelensure` (this is the default in `ini`-based languages).

EXAMPLE When creating a locale from another locale, you may want to reset some properties, like the BCP 47 tags. Here is an example of how to do it:

```
\babelprovide{spanish}      % import=es by default
\babelprovide[
    import=es,
    identification/tag.bcp47 = es-x-medieval,
    identification/extension.x.tag.bcp47 = medieval]
  {medievalspanish}
```

* * *

A few options (only luatex) set some properties of the writing system used by the language. These properties are *always* applied to the script, no matter which language is active. Although somewhat inconsistent, this makes setting a language up easier in most typical cases.

`onchar=ids | fonts | letters`

3.38 ※ This option is much like an 'event' called when a character belonging to the script of this locale is found (as its name implies, it acts on characters, not on spaces).

This option is particularly useful in pure Unicode multilingual text, because you can intermingle with no explicit markup differents scripts, each of which with its own hyphenation rules and font, and even its characteristic writing direction.

There are currently 3 'actions', which can be used at the same time (separated by spaces):

- With `ids`, both `\language` and `\localeid` are set to the values associated with this locale, affecting the hyphenation rules, as well as any transforms declared for it.
- With `fonts`, the fonts are switched to those specified for this locale (as set with `\babelfont`). Characters assigned to a locale can be customized with `\babelcharproperty`.
- 3.81 ※ Option `letters` restricts the 'actions' to letters, in the TEX sense (i.e., with catcode 11). Digits and punctuation are then considered part of current locale (as set by a selector). This option is useful when the main script is non-Latin and there is a secondary one whose script is Latin.

NOTE An alternative approach with luatex and Harfbuzz is the font option `RawFeature={multiscript=auto}`. It does not switch the babel language and therefore the line breaking rules, but in many cases it can be enough.

NOTE There is no general rule to set the font for a punctuation mark, because it is a semantic decision and not a typographical one. Consider the following sentence: "یک, دو, and سه are Persian numbers". In this case the punctuation font must be the English one, even if the commas are surrounded by non-Latin letters. Quotation marks, parenthesis, etc., are even more complex. Several criteria are possible, like the main language (the default in babel), the first letter in the paragraph, or the surrounding letters, among others, but even so manual switching can still be necessary.

5 Locale features

5.1 Hyphenation and line breaking – 1. Commands

Babel deals with three kinds of line breaking rules: Western, typically the LGC group, South East Asian, like Thai, and CJK, but support depends on the engine: pdftex only deals with the former, xetex also with the second one (although in a limited way) and CJK, while luatex provides basic rules for all of them, as well as additional rules (like Uyghur and Tibetan). With luatex there are also tools for non-standard hyphenation and line breaking rules, explained in the next section.

`\babelhyphenation[⟨language⟩,⟨language⟩,...]{⟨exceptions⟩}`

3.9a Sets hyphenation exceptions for the languages given or, without the optional argument, for *all* languages (e.g., proper nouns or common loan words, and of course monolingual documents). Multiple declarations work much like `\hyphenation` (last wins), but language exceptions take precedence over global ones.

It can be used only in the preamble, and exceptions are set when the language is first selected, thus taking into account changes of `\lccodes`'s done in `\extras`⟨*language*⟩ as well as the language-specific encoding (not set in the preamble by default). For example:

```
\babelhyphenation{Wal-hal-la Dar-bhan-ga}
```

Listed words are saved expanded and therefore it relies on the LICR. Of course, it also works without the LICR if the input and the font encodings are the same, like in Unicode based engines.

NOTE Using `\babelhyphenation` with Southeast Asian scripts is mostly pointless. But with `\babelpatterns` (below) you may fine-tune line breaking (only luatex). For example:

```
\babelpatterns[thai]{ศึก2ษา}
```

Even if there are no patterns for the language, you can add at least some typical cases.

NOTE Use `\babelhyphenation` instead of `\hyphenation` to set hyphenation exceptions in the preamble before any language is explicitly set with a selector. In the preamble the hyphenation rules are not always fully set up and an error can be raised.

`\babelpatterns[⟨language⟩,⟨language⟩,...]{⟨patterns⟩}`

3.9m *In luatex only*,[3] adds or replaces patterns for the languages given or, without the optional argument, for *all* languages. If a pattern for a certain combination already exists, it gets replaced by the new one.

It can be used only in the preamble, and patterns are added when the language is first selected, thus taking into account changes of `\lccodes`'s done in `\extras`⟨*language*⟩ as well as the language-specific encoding (not set in the preamble by default). Multiple `\babelpatterns`'s are allowed.

Listed patterns are saved expanded and therefore it relies on the LICR. Of course, it also works without the LICR if the input and the font encodings are the same, like in Unicode based engines.

3.31 (Only luatex.) With `\babelprovide` and `imported` CJK languages, a simple generic line breaking algorithm (push-out-first) is applied, based on a selection of the Unicode rules (3.32 it is disabled in verbatim mode, or more precisely when the hyphenrules are set to `nohyphenation`). It can be activated alternatively by setting explicitly the `intraspace`.

3.27 Interword spacing for Thai, Lao and Khmer is activated automatically if a language with one of those scripts is loaded with `\babelprovide`. See the sample on the babel repository. With both Unicode engines, spacing is based on the "current" em unit (the size of the previous char in luatex, and the font size set by the last `\selectfont` in xetex).

NOTE With Unicode engines, a line break can happen just before an explicit combining char (e.g., *g̃*, used in Guarani and Filipino, is not included as a combined char and it's represented in Unicode as `U+0067 U+0303`. This issue is not directly related to babel, but to the hyphenation patterns and/or the font renderer. However, at least with luatex there is a workaround (change the language name to what you are using):

```
\babelposthyphenation{guarani}{ | [{0300}-{036F}] }{ remove, {} }
```

The Lua pattern means 'a discretionary followed by a character in the range `U+0300`–`U+0367` (which contains combining chars)'. An alternative to a transform is `\babelpatterns`.

`\babelhyphenmins*[⟨language⟩,⟨language⟩,...]{⟨left⟩}{⟨right⟩}[⟨hyphenationmin⟩]`

24.10 ※ See the news page for the rationale for this command. It sets the corresponding values for the given languages (all languages without the optional argument). With the star, the values are also applied immediately (the optional argument and the star are currently incompatible). The optional argument is available only in luatex.

[3] With luatex exceptions and patterns can be modified almost freely. However, this is very likely a task for a separate package and `babel` only provides the most basic tools.

EXAMPLE You are typesetting a book with wide lines and want to limit the number of hyphens in all languages:

```
\babelhyphenmins{3}{4}
```

But there is also some 3-column text and you want to be more flexible:

```
\begin{multicols}{3}
\babelhyphenmins*{2}{3}
...
\end{multicols}
```

`\babelhyphen*{⟨type⟩}`
`\babelhyphen*{⟨text⟩}`

3.9a It is customary to classify hyphens in two types: (1) *explicit* or *hard hyphens*, which in TEX are entered as `-`, and (2) *optional* or *soft hyphens*, which are entered as `\-`. Strictly, a *soft hyphen* is not a hyphen, but just a breaking opportunity or, in TEX terms, a "discretionary"; a *hard hyphen* is a hyphen with a breaking opportunity after it. A further type is a *non-breaking hyphen*, a hyphen without a breaking opportunity.

In TEX, `-` and `\-` forbid further breaking opportunities in the word. This is the desired behavior very often, but not always, and therefore many languages provide shorthands for these cases. Unfortunately, this has not been done consistently: for example, `"-` in Dutch, Portuguese, Catalan or Danish is a hard hyphen, while in German, Spanish, Norwegian, Slovak or Russian it is a soft hyphen. Furthermore, some of them even redefine `\-`, so that you cannot insert a soft hyphen without breaking opportunities in the rest of the word.

Therefore, some macros are provided with a set of basic "hyphens" which can be used by themselves, to define a user shorthand, or even in language files.

- `\babelhyphen{soft}` and `\babelhyphen{hard}` are self explanatory.
- `\babelhyphen{repeat}` inserts a hard hyphen which is repeated at the beginning of the next line, as done in languages like Polish, Portuguese and Spanish.
- `\babelhyphen{nobreak}` inserts a hard hyphen without a break after it (even if a space follows).
- `\babelhyphen{empty}` inserts a break opportunity without a hyphen at all.
- `\babelhyphen{⟨text⟩}` is a hard "hyphen" using ⟨*text*⟩ instead. A typical case is `\babelhyphen{/}`.

With all of them, hyphenation in the rest of the word is enabled. If you don't want to enable it, there is a starred counterpart: `\babelhyphen*{soft}` (which in most cases is equivalent to the original `\-`), `\babelhyphen*{hard}`, etc.

Note `hard` is also good for isolated prefixes (e.g., *anti-*) and `nobreak` for isolated suffixes (e.g., *-ism*), but in both cases `\babelhyphen*{nobreak}` is usually better.

There are also some differences with LATEX: (1) the character used is that set for the current font, while in LATEX it is hardwired to `-` (a typical value); (2) the hyphen to be used in fonts with a negative `\hyphenchar` is `-`, as in LATEX, but it can be changed to another value by redefining `\babelnullhyphen`; (3) a break after the hyphen is forbidden if preceded by a glue >0 pt (at the beginning of a word, provided it is not immediately preceded by, say, a parenthesis).

It works with Plain.

`\begin{hyphenrules}{⟨language⟩} ... \end{hyphenrules}`

The environment `hyphenrules` can be used to select *only* the hyphenation rules to be used (it can be used as command, too). This can for instance be used to select 'nohyphenation', provided that in `language.dat` the 'language' nohyphenation is defined

by loading `zerohyph.tex`. It deactivates language shorthands, too (but not user shorthands).

Except for these simple uses, `hyphenrules` is deprecated and `otherlanguage*` (the starred version) is preferred, because the former does not take into account possible changes in encodings of characters like, say, `'` done by some languages (e.g., italian, french, ukrainian).

NOTE For the hyphenation to work correctly, lccodes cannot change, because TEX only takes into account the values when the paragraph is hyphenated, i.e., when it has been finished.[4] So, if you write a chunk of French text with `\foreignlanguage`, the apostrophes might not be taken into account. This is a limitation of TEX, not of babel. Alternatively, you may use `\useshorthands` to activate `'` and `\defineshorthand`, or redefine `\textquoteright` (the latter is called by the non-ASCII right quote).

5.2 Hyphenation and line breaking – 2. 'Provide' options

`justification=unhyphenated | kashida | elongated | padding`

3.59 ※ There are currently 4 options. Note they are language dependent, so that they will not be applied to other languages.

The first one (`unhyphenated`) activates a line breaking mode that allows breaks only at spaces, which can be stretched to arbitrary amounts. Although for European standards the result may look odd, in some writing systems, like Malayalam and other Indic scripts, this has been the customary (although not always the desired) practice. Because of that, no locale enables currently this mode by default (Amharic is an exception). Unlike `\sloppy`, the `\hfuzz` and the `\vfuzz` are not changed, because this line breaking mode is not really 'sloppy' (in other words, overfull boxes are reported as usual).

The second and the third are for the Arabic script. It sets the linebreaking and justification method, which can be based on the ARABIC TATWEEL character or in the 'justification alternatives' OpenType table (`jalt`). For an explanation see the babel site.

3.81 ※ The option `padding` has been devised primarily for Tibetan. It's still somewhat experimental. Again, there is an explanation in the babel site.

Hebrew justification based on interletter spacing is handled with a transform and doesn't require a specific justification method.

`linebreaking=`

3.59 ※ Just a synonymous for `justification`. Depending on the language, this name can make more sense.

`intraspace=`⟨*base*⟩ ⟨*shrink*⟩ ⟨*stretch*⟩

Sets the interword space for the writing system of the language, in em units (so, `0 .1 0` is `0em plus .1em`). Like `\spaceskip`, the em unit applied is that of the current text (more precisely, the previous glyph). Currently used only in Southeast Asian scripts, like Thai, and CJK.

`intrapenalty=`⟨*penalty*⟩

Sets the interword penalty for the writing system of this language. Currently used only in Southeast Asian scripts, like Thai. If 0, which is the default value, no penalty is inserted.

5.3 Shorthands – 1. Commands

A *shorthand* is a sequence of one or two characters that expands to arbitrary TEX code. Any language can have associated a set of shorthands, which is usually named as the language

[4] This explains why LATEX assumes the lowercase mapping of T1 and does not provide a tool for multiple mappings. Unfortunately, `\savinghyphcodes` is not a solution either, because lccodes for hyphenation are frozen in the format and cannot be changed.

which are intended for, and defined in the corresponding language definition file. Note, however, the name can be different.

In what follows, ⟨*shorthands-name*⟩ is the name of a set of shorthands. The name used is printed to the `log` file with something like:

```
Package babel Info: 'uppersorbian' activates 'usorbian' shorthands.
```

This is particularly true for “dialects”, because a single set of shorthands is re-used – an example is `austrian`, which activates the `german` shorthands.

Shorthands can be used for different kinds of things; for example: (1) in some languages shorthands such as `"a` are defined to be able to hyphenate the word if the encoding is `OT1`; (2) in some languages shorthands such as `!` are used to insert the right amount of white space; (3) several kinds of discretionaries and breaks can be inserted easily with `"-`, `"=`, etc.

The package inputenc as well as luatex and xetex have alleviated entering non-ASCII characters, but minority languages and some kinds of text can still require characters not directly available on the keyboards (and sometimes not even as separated or precomposed Unicode characters). As to the point 2, now pdfTeX provides `\knbccode`, and luatex can manipulate the glyph list. Tools for point 3 can still be very useful in general.

There are four levels of shorthands; by order of precedence: *user*, *language* (which is what the description above refers to), *system*, and *language user*. In most cases, you will use only shorthands provided by languages.

NOTE Keep in mind the following:

1. Activated chars used for two-char shorthands cannot be followed by a closing brace `}` and the spaces following are gobbled. With one-char shorthands (e.g., `:` in `french`), they are preserved.
2. If on a certain level (system, language, user, language user) there is a one-char shorthand, two-char ones starting with that char and on the same level are ignored.
3. Since they are active, a shorthand cannot contain the same character in its definition (except if deactivated with, e.g., `\string`).

TROUBLESHOOTING A typical error when using shorthands is the following:

```
! Argument of \language@active@arg" has an extra }.
```

It means there is a closing brace just after a shorthand, which is not allowed (e.g., `"}`). Just add `{}` after (e.g., `"{}}`).

`\shorthandon`{⟨*shorthands-list*⟩}

`\shorthandoff*`{⟨*shorthands-list*⟩}

It is sometimes necessary to switch a shorthand character off temporarily, because it must be used in an entirely different way. For this purpose, the user commands `\shorthandoff` and `\shorthandon` are provided. They each take a list of characters as their arguments.

The command `\shorthandoff` sets the `\catcode` for each of the characters in its argument to other (12); the command `\shorthandon` sets the `\catcode` to active (13). Both commands work on only ‘known’ shorthand characters, and an error will be raised otherwise. You can check if a character is a shorthand with `\ifbabelshorthand` (see below).

3.9a However, `\shorthandoff` does not behave as you would expect with characters like ~ or ^, because they usually are not “other”. For them `\shorthandoff*` is provided, so that with

```
\shorthandoff*{~^}
```

~ is still active, very likely with the meaning of a non-breaking space, and ^ is the superscript character. The catcodes used are those when the shorthands are defined, usually when language files are loaded.

If you do not need shorthands, or prefer an alternative approach of your own, you may want to switch them off with the package option `shorthands=off`, as described below.

WARNING It is worth emphasizing that these macros are meant for temporary changes. Whenever possible, shorthands must be always enabled or disabled.

`\useshorthands*{⟨char⟩}`

It initiates the definition of user-defined shorthand sequences. It has one argument, the character that starts these personal shorthands.

3.9a User shorthands are not always alive, as they may be deactivated by languages (for example, if you use " for your user shorthands and switch from german to french, they stop working). Therefore, a starred version `\useshorthands*{⟨char⟩}` is provided, which makes sure shorthands are always activated.

If the package option `shorthands` is used, you must include any character to be activated with `\useshorthands`.

`\defineshorthand[⟨shorthands-name⟩,⟨shorthands-name⟩...]{⟨shorthand⟩}{⟨code⟩}`

It takes two arguments: the first is a one- or two-character shorthand sequence, and the second is the code the shorthand should expand to.

3.9a An optional argument allows to (re)define language and system shorthands (some languages do not activate shorthands, so you may want to add `\languageshorthands{⟨shorthands-name⟩}` to the corresponding `\extras⟨language⟩`, as explained below). By default, user shorthands are (re)defined.

User shorthands override language ones, which in turn override system shorthands. Language-dependent user shorthands take precedence over “normal” user shorthands.

EXAMPLE Let’s assume you want a unified set of shorthands for discretionaries (languages do not define shorthands consistently, and `"-`, `\-`, `"=` have different meanings). You can start with, say:

```
\useshorthands*{"}
\defineshorthand{"*}{\babelhyphen{soft}}
\defineshorthand{"-}{\babelhyphen{hard}}
```

However, the behavior of hyphens is language-dependent. For example, in languages like Polish and Portuguese, a hard hyphen inside compound words are repeated at the beginning of the next line. You can then set:

```
\defineshorthand[*polish,*portuguese]{"-}{\babelhyphen{repeat}}
```

Here, options with * set a language-dependent user shorthand, which means the generic declarations above for `"-` apply to all languages *except* `polish` and `portuguese`; without * they would (re)define the language shorthands instead, which are overridden by user ones.

Now, you have a single unified shorthand (`"-`), with a content-based meaning (‘compound word hyphen’) whose visual behavior is that expected in each context.

`\languageshorthands{⟨shorthands-name⟩}`

Used to switch the shorthands on the language level. It takes one argument, the name of a language (more precisely, the name assigned to ⟨*shorthands-name*⟩ by the language) or `none` (the latter does what its name suggests).[5]

[5] Actually, any name not corresponding to a language group does the same as `none`. However, follow this convention because it might be enforced in future releases of babel to catch possible errors.

Note that for this to work the language should have been specified as an option when loading the babel package. For example, by also loading `german` you can use in english the shorthands defined by the former with

```
\addto\extrasenglish{\languageshorthands{german}}
```

(You may also need to activate them as user shorthands in the preamble with, for example, `\useshorthands` or `\useshorthands*`.)

EXAMPLE Very often, this is a more convenient way to deactivate shorthands than `\shorthandoff`, for example if you want to define a macro to ease typing phonetic characters with tipa:

```
\newcommand{\myipa}[1]{{\languageshorthands{none}\tipaencoding#1}}
```

`\babelshorthand{`⟨*shorthand*⟩`}`

With this command you can use a shorthand even if (1) not activated in `shorthands` (in this case only shorthands for the current language are taken into account, i.e., not user shorthands), (2) turned off with `\shorthandoff` or (3) deactivated with the internal `\bbl@deactivate`; for example, `\babelshorthand{"u}` or `\babelshorthand{:}`. (You can conveniently define your own macros, or even your own user shorthands provided they do not overlap.)

EXAMPLE Since by default shorthands are not activated until `\begin{document}`, you may use this macro when defining the `\title` in the preamble:

```
\title{Documento científico\babelshorthand{"-}técnico}
```

For your records, here is a list of shorthands, but you must double check them, as they may change:[6]

Languages with no shorthands Croatian, English (any variety), Indonesian, Hebrew, Interlingua, Irish, Lower Sorbian, Malaysian, North Sami, Romanian, Scottish, Welsh

Languages with only " as defined shorthand character Albanian, Bulgarian, Danish, Dutch, Finnish, German (old and new orthography, also Austrian), Icelandic, Italian, Norwegian, Polish, Portuguese (also Brazilian), Russian, Serbian (with Latin script), Slovene, Swedish, Ukrainian, Upper Sorbian

Basque " ' ~
Breton : ; ? !
Catalan " ' `
Czech " -
Esperanto ^
Estonian " ~
French (all varieties) : ; ? !
Galician " . ' ~ ⟨ ⟩
Greek (ancient, polytonic, only 8-bit TEX) ~, (optional, see the manual for Greek) ;
Hungarian `
Kurmanji ^
Latin " ^ ' =
Slovak " ^ ' -
Spanish " . ⟨ ⟩ ' ~
Turkish : ! =

[6] Thanks to Enrico Gregorio.

In addition, the babel core declares ~ as a one-char shorthand which is let, like the standard ~, to a non breaking space. Actually, this declaration serves to nothing, but it is preserved for backward compatibility.

`\ifbabelshorthand{⟨character⟩}{⟨true⟩}{⟨false⟩}`

3.23 Tests if a character has been made a shorthand.

NOTE Both ltxdoc and babel use `\AtBeginDocument` to change some catcodes, and babel reloads hhline to make sure `:` has the right one, so if you want to change the catcode of `|` it has to be done using the same method at the proper place, with

```
\AtBeginDocument{\DeleteShortVerb{\|}}
```

before loading babel. This way, when the document begins the sequence is (1) make `|` active (ltxdoc); (2) make it inactive (your settings); (3) make babel shorthands active (babel); (4) reload hhline (babel, now with the correct catcodes for `|` and `:`).

NOTE Using a character mathematically active (i.e., with math code `"8000`) as a shorthand can make TEX enter in an infinite loop in some rare cases. (There is a partial solution.)

5.4 Shorthands – 2. Package options

`activeacute`

`activegrave`

For some languages babel supports these options to set ' and `, respectively, as a shorthand in case it is not done by default.

`shorthands=⟨char⟩⟨char⟩... | off`

The only language shorthands activated are those given, like, e.g.:

```
\usepackage[esperanto,french,shorthands=:;!?]{babel}
```

If ' is included, `activeacute` is set; if ` is included, `activegrave` is set. Active characters (like ~) should be preceded by `\string` (otherwise they will be expanded by LATEX before they are passed to the package and therefore they will not be recognized); however, `t` is provided for the common case of ~, and `c` for the not so common case of the comma.

With `shorthands=off` no language shorthands are defined. As some languages use this mechanism for tools not available otherwise, a macro `\babelshorthand` is defined, which allows using them; see above.

`safe=none | ref | bib`

Some LATEX macros are redefined so that using shorthands is safe. With `safe=bib` only `\nocite`, `\bibcite` and `\bibitem` are redefined. With `safe=ref` only `\newlabel`, `\ref` and `\pageref` are redefined (as well as a few macros from varioref and ifthen).

With `safe=none` no macro is redefined. This option is strongly recommended, because a good deal of incompatibilities and errors are related to these redefinitions. As of 3.34 , in ε-TEX based engines (i.e., almost every engine except the oldest ones) shorthands can be used in these macros (formerly you could not).

`math=active | normal`

Shorthands are mainly intended for text, not for math. By setting this option with the value `normal` they are deactivated in math mode (default is `active`) and things like `${a'}$` (a closing brace after a shorthand) are not a source of trouble anymore.

5.5 Digits and counters

3.20 About thirty `ini` files define a field named `digits.native`. When it is present, two macros are created: \⟨*language*⟩`digits` and \⟨*language*⟩`counter` (only luatex and xetex). With the first, a string of 'Latin' digits are converted to the native digits of that language; the second takes a counter name as argument. With the option `maparabic` in `\babelprovide`, `\arabic` is redefined to produce the native digits.

NOTE `maparabic` redefines `\arabic` *globally*, to avoid inconsistencies in, for example, page numbering; note as well dates do not rely on `\arabic`.

For example:

```
\usepackage[telugu]{babel}
  % Or also, if you want, with:
  % provide={ maparabic }
\babelfont{rm}{Gautami}
\begin{document}
\telugudigits{1234}
\telugucounter{section}
\end{document}
```

Languages providing native digits in all or some variants are:

Arabic	Dzongkha	Lao	Odia	Thai
Assamese	Gujarati	Maithili	Pashto	Tibetan
Bangla	Haryanvi	Malayalam	Persian	Urdu
Bhojpuri	Hindi	Manipuri	Punjabi	Uyghur
Bodo	Hmong Njua	Marathi	Rajasthani	Uzbek
Burmese	Kannada	Mazanderani	Sanskrit	Vai
Cantonese	Kashmiri	Nepali	Santali	
Central Kurdish	Khmer	Northern	Sindhi	
Chinese	Konkani	Kurdish	Tamil	
Dogri	Korean	Northern Luri	Telugu	

3.30 With luatex there is an alternative approach for mapping digits, namely, `mapdigits`. Conversion is based on the language and it is applied to the typeset text (not math, PDF bookmarks, etc.) before bidi and fonts are processed (i.e., to the node list as generated by the TEX code). This means the local digits have the correct bidirectional behavior (unlike `Numbers=Arabic` in fontspec, which is deprecated).

Another option is the transform `digits.native` (see 5.7), or the `digits` style with `\localenumeral` and `\localecounter` (see below).

NOTE With xetex you can use the option `Mapping` when defining a font.

`\localenumeral{`⟨*style*⟩`}{`⟨*number*⟩`}`
`\localecounter{`⟨*style*⟩`}{`⟨*counter*⟩`}`

3.41 ※ Many `ini` locale files provide information about non-positional numerical systems, based on those predefined in CSS. They only work with luatex and xetex and are fully expandable (even inside an unprotected `\edef`). Currently, they are limited to numbers below 10000.

There are several ways to use them (for the available styles in each language, see the list below):

- `\localenumeral{`⟨*style*⟩`}{`⟨*number*⟩`}`, like `\localenumeral{abjad}{15}`
- `\localecounter{`⟨*style*⟩`}{`⟨*counter*⟩`}`, like `\localecounter{lower}{section}`
- In `\babelprovide`, as an argument to the keys `alph` and `Alph`, which redefine what `\alph` and `\Alph` print. For example:

```
\babelprovide[alph=alphabetic]{thai}
```

The styles are (see also *List of counters*):

Ancient Greek `lower.ancient`, `upper.ancient`
Amharic `afar`, `agaw`, `ari`, `blin`, `dizi`, `gedeo`, `gumuz`, `hadiyya`, `harari`, `kaffa`, `kebena`, `kembata`, `konso`, `kunama`, `meen`, `oromo`, `saho`, `sidama`, `silti`, `tigre`, `wolaita`, `yemsa`
Arabic `abjad`, `maghrebi.abjad`
Armenian `lower.letter`, `upper.letter`
Belarusan, Bulgarian, Church Slavic, Macedonian, Serbian `lower`, `upper`
Bangla `alphabetic`
Central Kurdish `alphabetic`
Chinese `cjk-earthly-branch`, `cjk-heavenly-stem`, `circled.ideograph`, `parenthesized.ideograph`, `fullwidth.lower.alpha`, `fullwidth.upper.alpha`, `formal`, `informal`.
Church Slavic (Glagolitic) `letters`
Coptic `epact`, `lower.letters`
French `date.day` (mainly for internal use).
Georgian `letters`
Greek `lower.modern`, `upper.modern`, `lower.ancient`, `upper.ancient` (all with keraia), `lower.modern.nonumeralsign`, `upper.modern.nonumeralsign` (without keraia).
Hebrew `letters` (3.93※ if the language is loaded explicitly, also `letters.plain`, `letters.gershayim`, `letters.final`)
Hindi `alphabetic`
Italian `lower.legal`, `upper.legal`
Japanese `hiragana`, `hiragana.iroha`, `katakana`, `katakana.iroha`, `circled.katakana`, `informal`, `formal`, `cjk-earthly-branch`, `cjk-heavenly-stem`, `circled.ideograph`, `parenthesized.ideograph`, `fullwidth.lower.alpha`, `fullwidth.upper.alpha`
Khmer `consonant`
Korean `consonant`, `syllable`, `hanja.informal`, `hanja.formal`, `hangul.formal`, `cjk-earthly-branch`, `cjk-heavenly-stem`, `circled.ideograph`, `parenthesized.ideograph`, `fullwidth.lower.alpha`, `fullwidth.upper.alpha`
Marathi `alphabetic`
Persian `abjad`, `alphabetic`
Russian `lower`, `lower.full`, `upper`, `upper.full`
Syriac `letters`
Tamil `ancient`
Thai `alphabetic`
Ukrainian `lower` , `lower.full`, `upper` , `upper.full`

3.45※ In addition, native digits (in languages defining them) may be printed with the numeral style `digits`.

25.12※ With `mapdot=`⟨*text*⟩ as an option in 'provide', the standard counters (those starting with `\the...`) are traversed to replace each dot by the provided ⟨*text*⟩ in a locale-dependent way. Without a value, a default is used — currenty the only default value is for the Arabic (`ar`) locales, which is set to a hyphen (because the zero and the dot are very similar in this script). See the news page for further details.

5.6 Dates

3.45※ When the data is taken from an `ini` file, you may print the date corresponding to the Gregorian calendar and other lunisolar systems with the following command.

`\localedate`[⟨*calendar=.., variant=.., convert[=..]*⟩]{⟨*year*⟩}{⟨*month*⟩}{⟨*day*⟩}

By default the calendar is the Gregorian, but an `ini` file may define strings for other calendars: `am` (`ethiopic`), `ar` and `ar-*` (`islamic`), `cop` (`coptic`), `fa` (`islamic`, `persian`), `he` (`hebrew`), `hi` (`indian`), `th` (`buddhist`), and (25.16※) `el` (`julian`). In the latter case, the three arguments are the year, the month, and the day for those in the corresponding calendar.

They are *not* the Gregorian date to be converted (which means, say, 13 is a valid month number with `calendar=hebrew` and `calendar=coptic`). However, with the option `convert` it's converted (using internally the command described next).

Even with a certain calendar there may be variants. In Kurdish the default variant prints something like *30. Çileya Pêşîn 2019*, but with `variant=izafa` it prints *31'ê Çileya Pêşînê 2019*.

The option `convert` can take a value for an external converter, with the format ⟨*provider*⟩:⟨*calendar*⟩; currently there is just a provider named *calendrica*, and only for luatex (see 26.10※ for details).

The default calendar for a language can be set in `\babelprovide`, with the key `calendar` (an empty value is the same as `gregorian`). In this case, `\today` always converts the date. Variants are preceded by a dot, so that `calendar = .genitive` in `serbian` `\today` selects the date in this variant (more explicitly is `gregorian.genitive`). As a last element, separated with a dot if there is a calendar or a variant, you can request a conversion with CALENDRICA.

EXAMPLE By default, `thai` prints the date with `\today` in the Buddhist calendar, but if you need a date in the Gregorian one, write:

```
\localedate[calendar=gregorian]{\year}{\month}{\day}
```

(Remember `\year`, `\month` and `\day` is the current Gregorian date, so no conversion is necessary.)

EXAMPLE On the other hand (and following the CLDR), the preferred calendar in most locales for Arabic is `gregorian` (in `ar-SA` is `islamic-umalqura`), so to set `islamic-civil` as the default one:

```
\babelprovide[import, calendar=islamic-civil]{arabic}
```

EXAMPLE The traditional `hindu-solar` calendar with `indian` strings, set for `kannada` as a package option:

```
\usepackage[
  kannada,
  provide={ calendar=indian.calendrica:hindu-solar }]
{babel}
```

`\babelcalendar`[⟨*date*⟩]{⟨*calendar*⟩}{⟨*year-macro*⟩}⟨*month-macro*⟩⟨*day-macro*⟩

3.76※ Although calendars aren't the primary concern of babel, the package should be able to, at least, generate correctly the current date in the way users would expect in their own culture. Currently, `\localedate` can print dates in a few calendars (provided the ini locale file has been imported), but year, month and day had to be entered by hand, which is inconvenient. With this macro, the current date is converted and stored in the three last arguments, which must be macros. Allowed calendars are:

`buddhist`	`ethiopic`	`islamic-civil`	`persian`
`chinese` 3.94※	`hebrew`	`islamic-umalqura`	
`coptic`	`indian` 26.11※	`julian`	

The optional argument converts the given date, in the form '⟨*year*⟩-⟨*month*⟩-⟨*day*⟩', although for practical reasons most calendars accept only a restricted range of years. Please, refer to the page on the news for 3.76 in the babel site for further details.

5.7 Transforms

Transforms (only luatex) provide a way to process the text on the typesetting level in several language-dependent ways, like non-standard hyphenation, special line breaking rules, script to script conversion, spacing conventions and so on.

NOTE They are similar in concept, but not the same, as those in Unicode. Actually, the main inspiration for this feature is the Omega transformation processes.

It currently embraces `\babelprehyphenation` and `\babelposthyphenation`.

3.57 ※ Several ini files predefine some transforms. They are activated with the key `transforms` in `\babelprovide`, either if the locale is being defined with this macro or the languages has been previously loaded as a class or package option, as the following example illustrates:

```
\usepackage[hungarian]{babel}
\babelprovide[transforms = digraphs.hyphen]{hungarian}
```

3.67 ※ Transforms predefined in the `ini` locale files can be made attribute-dependent, too. When an attribute between parenthesis is inserted subsequent transforms will be assigned to it (up to the list end or another attribute). For example, and provided an attribute called `\withsigmafinal` has been declared:

```
transforms = transliteration.omega (\withsigmafinal) sigma.final
```

This applies `transliteration.omega` always, but `sigma.final` only when `\withsigmafinal` is set.

Here are the transforms currently predefined. (A few may still require some fine-tuning. More to follow in future releases.)

	`digits.native`	An alternative to `mapdigits`, available in the same locales. This transform is applied before the first prehyphenation, while `mapdigits` is applied after the last posthyphenation. Another difference is `mapdigits` cannot be disabled in the middle of a paragraph. (This transform is *not* declared explicitly in `ini` files. Instead, it's defined by `babel` if the key `numbers/digits.native` exists.) 24.9 ※
Arabic	`transliteration.dad`	Applies the transliteration system devised by Yannis Haralambous for dad (simple and TEX-friendly). Not yet complete, but sufficient for most texts.
Arabic, Persian	`kashida.base`	Much like the following, but with diacritics stacked in the actual base character and not the kashida extension. With evenly inserted tatweels results are better. 3.94 ※
Arabic, Persian	`kashida.plain`	A very simple and basic transform for 'plain' Arabic fonts, which attempts to distribute the tatweel as evenly as possible (starting at the end of the line). See the news for version 3.59.
Chinese	`tone.numeric`	With `chinese-pinyin`, numeric tone marks are converted to diacritics. For example, `he2ping2` outputs *héping*. Valid marks are assumed. 26.12 ※
Chinese, Japanese	`input.nospaces`	With it, and just for convenience, spaces (and new lines) in the input are ignored.

Chinese, Japanese	`spacing.basic`	Basic rules for readjusting spacing. See 25.6 ※ for further details.
Croatian	`digraphs.ligatures`	Ligatures *DŽ*, *Dž*, *dž*, *LJ*, *Lj*, *lj*, *NJ*, *Nj*, *nj*. It assumes they exist. This is not the recommended way to make these transformations (the best way is with OTF features), but it can get you out in a hurry.
Croatian, Czech, Polish, Portuguese, Slovak, Spanish	`hyphen.repeat`	Explicit hyphens behave like `\babelhyphen {repeat}`.
Czech, Polish, Slovak	`oneletter.nobreak`	Converts a space after a non-syllabic preposition or conjunction into a non-breaking space.
Dutch	`diaeresis.hyphen`	Removes the diaeresis above a vowel if hyphenated just before.
Finnish	`prehyphen.nobreak`	Line breaks just after hyphens prepended to words are prevented, as in “pakastekaapit ja -arkut”.
French	`punctuation.space`	Rules for proper spacing with characters *;:!?«»* are applied.
German	`longs.unifraktur`	Implements the basic heuristic rules for the long s (ſ) from those in Unifraktur Maguntia. Although discretionaries aren’t taken into account, the transform is declared in the posthyphenation group, to ease if necessary fine tuning the rules for, e.g., prefixes and compound words. They are available in all German locales. See 25.5 ※ for further details and an example.
Greek	`diaeresis.hyphen`	Removes the diaeresis above iota and upsilon if hyphenated just before. It works with the three variants.
Greek	`transliteration.omega`	Although the provided combinations are not the full set, this transform follows the syntax of Omega: = for the circumflex, v for digamma, and so on. For better compatibility with Levy’s system, ~ (as ‘string’) is an alternative to =. ' is tonos in Monotonic Greek, but oxia in Polytonic and Ancient Greek.
Greek	`sigma.final`	The transliteration system above does not convert the sigma at the end of a word (on purpose). This transform does it. To prevent the conversion (an abbreviation, for example), write `"s`.
Hebrew, Yiddish, Ladino, Ancient Hebrew	`transliteration.cj`	A transliteration system based on that devised by Christian Justen for ‘cjhebrew‘. Final letters are not converted, and the furtive patah is not shifted.

Hebrew, Yiddish, Ladino, Ancient Hebrew	`justification.interletter`	Hebrew justification is based on variations in the spacing between individual letters within words. This transform activates this justification method. See 25.8 ※ for further details and an example.
Hindi, Sanskrit	`punctuation.space`	Inserts a space before the following four characters: *!?:;* .
Hindi, Sanskrit	`transliteration.hk`	The Harvard-Kyoto system to romanize Devanagari.
Hindi, Sanskrit	`transliteration.iast`	The IAST system to romanize Devanagari (thanks to Maximilian Mehner).
Hungarian	`digraphs.hyphen`	Hyphenates the long digraphs *ccs*, *ddz*, *ggy*, *lly*, *nny*, *ssz*, *tty* and *zzs* as *cs-cs*, *dz-dz*, etc.
Indic scripts	`danda.nobreak`	Prevents a line break before a danda or double danda if there is a space. For Assamese, Bangla, Gujarati, Hindi, Kannada, Malayalam, Marathi, Odia, Tamil, Telugu.
Japanese	`linebreak.strict`	Prevents line breaks before small kana variants.
Latin	`digraphs.ligatures`	Replaces the groups *ae*, *AE*, *oe*, *OE* with *æ*, *Æ*, *œ*, *Œ*.
Latin	`letters.noj`	Replaces *j*, *J* with *i*, *I*.
Latin	`letters.uv`	Replaces *v*, *U* with *u*, *V*.
Russian	`transliteration.gost779b`	The GOST 7.79-2000 System B for the romanization of Russian. 25.10 ※
Serbian	`transliteration.gajica`	(Note `serbian` with `ini` files refers to the Cyrillic script, which is here the target.) The standard system devised by Ljudevit Gaj.

`\babelposthyphenation`[⟨*options*⟩]{⟨*hyphenrules-name*⟩}{⟨*lua-pattern*⟩}{⟨*replacement*⟩}

3.37-3.39 ※ *With luatex* it is possible to define non-standard hyphenation rules, like `f-f` ▸ `ff-f`, repeated hyphens, ranked ruled (or more precisely, 'penalized' hyphenation points), and so on. A few rules are currently provided (see above), but they can be defined as shown in the following example, where `{1}` is the first captured char (between `()` in the pattern):

```
\babelposthyphenation{german}{([fmtrp]) | {1}}
{
  { no = {1}, pre = {1}{1}- }, % Replace first char with disc
  remove,                      % Remove automatic disc (2nd node)
  {}                           % Keep last char, untouched
}
```

In the replacements, a captured char may be mapped to another, too. For example, if the first capture reads `([íűǘ])`, the replacement could be `{1|íű|íú}`, which maps *íˊ* to *í*, and *ǘ* to *ú*, so that the diaeresis is removed.

This feature is activated with the first `\babelposthyphenation` or `\babelprehyphenation`.

3.67 ※ With the optional argument you can associate a user-defined transform to an attribute, so that it's active only when it's set (currently its attribute value is ignored). With this mechanism transforms can be set or unset even in the middle of paragraphs, and applied to single words. To define, set and unset the attribute, the LaTeX kernel provides the macros `\newattribute`, `\setattribute` and `\unsetattribute`. The following example shows how to use it, provided an attribute named `\latinnoj` has been declared:

```
\babelprehyphenation[attribute=\latinnoj]{latin}{ J }{ string = I }
```

See the babel site for a more detailed description and some examples. It also describes a few additional replacement types (`string`, `penalty`).

3.85 ※ Another option is `label`, which takes a value similar to those in `\babelprovide` key `transforms` (in fact, the latter just applies this option). This label can be used to turn on and off transforms with a higher level interface, by means of `\enablelocaletransform` and `\disablelocaletransform` (see below).

3.85 ※ When used in conjunction with `label`, this key makes a transform font dependent. As an example, the rules for Arabic kashida can differ depending on the font design. The value consists in a list of space-separated font tags:

```
\babelprehyphenation[label=transform.name, fonts=rm sf]{..}{..}
```

Tags can adopt two forms: a family, such as `rm` or `tt`, or the set family/series/shape. If a font matches one of these conditions, the transform is enabled. The second tag in `rm rm/n/it` is therefore redundant. There are no wildcards; so, for italics you may want to write something like `sf/m/it sf/b/it`.

Transforms set for specific fonts (at least once in any language) are always reset with a font selector.

In `\babelprovide`, transform labels can be tagged before their name, with a list separated with colons, like:

```
transforms = rm:sf:transform.name
```

Tranforms are executed in the same order they are declared. 25.9 ※ They can be also added at the beginning with the key `prepend`, which is particularly useful with predefined transforms. For example:

```
\babelprehyphenation[prepend]{hebrew}{-}{ string = ־ } % hyphen to maqaf
```

Although the main purpose of post-hyphenation transforms was non-standard hyphenation, they may actually be used for other transformations (for example, rules for long s in German).

`\babelprehyphenation`[⟨*options*⟩]{⟨*locale-name*⟩}{⟨*lua-pattern*⟩}{⟨*replacement*⟩}

3.44-3-52 ※ It is similar to the latter, but (as its name implies) applied before hyphenation, which is particularly useful in transliterations. There are other differences: (1) the first argument is the locale instead of the name of the hyphenation patterns; (2) in the search patterns = has no special meaning, while | stands for an ordinary space; (3) in the replacement, discretionaries are not accepted.

See the description above for the optional argument.

This feature is activated with the first `\babelposthyphenation` or `\babelprehyphenation`.

EXAMPLE You can replace a character (or series of them) by another character (or series of them). Thus, to enter *ž* as `zh` and *š* as `sh` in a newly created locale for transliterated Russian:

```
\babelprovide[hyphenrules=+]{russian-latin}   % Create locale
\babelprehyphenation{russian-latin}{([sz])h}  % Create rule
{
  string = {1|sz|šž},
  remove
}
```

EXAMPLE The following rule prevent the word “a” from being at the end of a line:

```
\babelprehyphenation{english}{|a|}
  {}, {},                       % Keep first space and a
  { insert, penalty = 10000 },  % Insert penalty
  {}                            % Keep last space
}
```

NOTE With luatex there is another approach to make text transformations, with the function `fonts.handlers.otf.addfeature`, which adds new features to an OTF font (substitution and positioning). These features can be made language-dependent, and babel by default recognizes this setting if the font has been declared with `\babelfont`. The *transforms* mechanism supplements rather than replaces OTF features.

With xetex, where *transforms* are not available, there is still another approach, with font mappings, mainly meant to perform encoding conversions and transliterations. Mappings, however, are linked to fonts, not to languages.

`\enablelocaletransform{`⟨*label*⟩`}`
`\disablelocaletransform{`⟨*label*⟩`}`

3.85 ※ Enables and disables the transform with the given label in the current language. Font dependent transforms are always enabled and cannot be disabled.

`\SetTrasformValue{`⟨*locale-name*⟩`}{`⟨*variable-name*⟩`}{`⟨*value*⟩`}`

This command allows adjustments (specifically numeric parameters in prehyphenation) in transform, which can be useful in those predeclared in `ini` files. Something like `{<variable-name>|<default>}` employs the variable as set by this macro. The following variables are currently defined:

- For French and `punctuation.space`: `guillemet.natural`, `guillemet.plus`, `guillemet.minus`, `colon.natural`, `colon.plus`, `colon.minus`, `high.natural`, `high.plus`, `high.minus`. A value of 1 is the current ordinary space. 24.13 ※
- For languages in the Hebrew script and `justification.interletter`: `interletter.plus`, which sets the factor for the glue 'plus' value. By default, .5. 25.8 ※
- For Japanese and `linebreak.strict`: `small.natural`, `small.plus` and `small.minus`. They are set when the transform is activated to the intraspace values. In addition, the penalty, by default 10000, can be set with `small.penalty`. 24.14 ※

EXAMPLE With:

```
\SetTransformValue{french}{colon.natural}{.8}
```

the natural spacing with the colon is made slightly smaller (default is 1).

`\ShowBabelTransforms{`⟨*string*⟩`}`

25.7 ※ Applies the current transforms to the string and shows in the `log` the transformations performed. Don't rely on the current format, because it may change. In the process, penalties, discretionaries, etc., can be inserted, which are currently printed as a boxed '?', as a boxed 'US' (because it's the 'unit separator') or in another way. With this macros you can better understand what's going on, so that you can debug the transforms. Use it only in the document body.

5.8 Support for xetex interchar

3.97 ※ A few macros are provided to deal with locale dependent inter-character rules (aka 'interchar').

`\babelcharclass{⟨locale⟩}{⟨name⟩}{⟨char-list⟩}`

Declares a new character class, which is assigned to the characters in `{⟨char-list⟩}`, entered either as characters or in macro form (e.g., `\}`). If you need to enter them by their numeric value, use the TEX ^-notation (e.g., `^^^^1fa0`). Ranges are allowed, with a hyphen (e.g., `.,;a-zA-Z`). If you need the hyphen to be assigned a class, write it at the very beginning of the list.

There are several predefined 'global' classes, namely `default`, `cjkideogram`, `cjkleftpunctuation`, `cjkrightpunctuation`, `boundary`, and `ignore`, which are described in the xetex manual. These classes are used by the `linebreak.basic`, described below.

`\babelinterchar[⟨options⟩]{⟨locale⟩} {⟨class-first⟩}{⟨class-second⟩}{⟨code⟩}`

`{⟨class-first⟩}` and `{⟨class-second⟩}` can be comma separated lists of char classes, and all combinations are defined (so that 2 first classes with 2 second classes, define 4 combinations). In the ⟨*options*⟩ field a key named `label` is available, which allows to enable or to disable the rule with the following two commands. Like prehyphenation transforms in luatex, interchars are not applied if the current hyphenation rules are `nohyphenation`.

`\enablelocaleinterchar{⟨label⟩}`
`\disablelocaleinterchar{⟨label⟩}`

Enable or disable the interchar rules with the given label for the current language.

EXAMPLE Not very useful, but illustrative (taken from the unfortunately obsolete interchar package, by Zou Ho), to colorize the letters 'x' and 'y' (this way to group text is usually not a good idea, however).

```
\usepackage{color}
\babelcharclass{english}{colored}{xy}
\babelinterchar{english}{default, boundary}{colored}{\bgroup\color{red}}
\babelinterchar{english}{colored}{default, boundary}{\egroup}
```

A more realistic example follows, which inserts a thin space between a digit and a percent sign. Note the former is entered as a range, and the latter in command form:

```
\babelcharclass{english}{digit}{0-9}
\babelcharclass{english}{percent}{\%}
\babelinterchar[label=percent]{english}{digit}{percent}{\,}
```

WARNING Keep in mind two points: (1) a character can be assigned a single class; this is a limitation in the interchar mechanisms that often leads to incompatibilities; (2) since the character classes set with `\babelcharclass` are saved (so that they can be restored), there is a limit in the number of characters in the `{⟨char-list⟩}` (which, however, must be large enough for many uses).

`interchar=⟨interchar-list⟩`

24.1 ※ This key in `\babelprovide` activates predefined rules for the 'provided' locale. Currently the following `interchar`'s are defined:

Cantonese, Chinese, Japanese, Korean	`linebreak.basic`	24.4 ※ Basic settings for CJK defined in (plain) xetex. See the linked news page for details.

French	`punctuation.space`	Rules for proper spacing with characters *;:!?«»* are applied.

WARNING This feature requires `import`.

NOTE You can use `transforms` and `interchar` at the same time. Only the relevant key for the current engine is taken into account.

5.9 Scripts

Babel provides no standard interface to select scripts, because they are best selected with languages tied to them. In pdftex, scripts are indirectly managed by means of the low-level `\fontencoding`, whose direct use is discouraged — even the Latin script may require different encodings (i.e., sets of glyphs) depending on the language, and therefore such a switch would be in a sense incomplete. Note the so-called Unicode fonts (in luatex and xetex) do not improve the situation either. So, a font suited for Vietnamese is not necessarily suited for, say, the romanization of Indic languages, and the fact it contains glyphs for Modern Greek does not mean it includes them for Ancient Greek.

NOTE Some languages sharing the same script define macros to switch it (e.g., `\textcyrillic`), but be aware they may also set the language to a certain default. Even the babel core defined `\textlatin`, but it was somewhat buggy because in some cases it messed up encodings and fonts (for example, if the main Latin encoding was `LY1`), and therefore it has been deprecated (but still defined for backwards compatibility).

The locales currently available cover the following scripts:

Arabic
Armenian
Avestan
Bangla
Batak
Bengali
Buginese
Carian
Chakma
Cherokee
Coptic
Cyrillic
Devanagari
Egyptian hieroglyphs
Ethiopic
Georgian
Glagolitic
Gothic
Greek
Gujarati
Gurmukhi
Han
Hebrew
Imperial Aramaic
Japanese
Kannada
Khmer
Khojki
Khudawadi
Korean
Lao
Latin
Lepcha
Limbu
Linear A
Lycian
Lydian
Malayalam
Mandaean
Myanmar
Nabataean
New Tai Lue
Nyiakeng Puachue Hmong
N'Ko
Ogham
Ol Chiki
Old Church Slavonic Cyrillic
Old Italic
Old Persian
Old South Arabian
Old Uyghur
Oriya
Osage
Palmyrene
Phoenician
Runic
Samaritan
Simplified
Sinhala
Sumero-Akkadian Cuneiform
Syriac
Tai Le
Tamil
Tangut
Telugu
Thaana
Thai
Tibetan
Tifinagh
Traditional
Ugaritic
Unified Canadian Aboriginal Syllabics
Vai
Yi

`\ensureascii{`⟨*text*⟩`}`

3.9i This macro makes sure ⟨*text*⟩ is typeset with a LICR-savvy encoding in the ASCII range. It is used to redefine `\TeX` and `\LaTeX` so that they are correctly typeset even with `LGR` or `X2` (the complete list is stored in `\BabelNonASCII`, which by default is `LGR`, `LGI`, `X2`, `OT2`, `OT3`, `OT6`, `LHE`, `LWN`, `LMA`, `LMC`, `LMS`, `LMU`, but you can modify it). So, in some sense it fixes the bug described in the previous paragraph.

If non-ASCII encodings are not loaded (or no encoding at all), it is no-op (also `\TeX` and `\LaTeX` are not redefined); otherwise, `\ensureascii` switches to the encoding at the beginning of the document if ASCII-savvy, or else the last ASCII-savvy encoding loaded. For

example, if you load LY1,LGR, then it is set to LY1, but if you load LY1,T2A it is set to T2A. The symbol encodings TS1, T3, and TS3 are not taken into account, since they are not used for "ordinary" text (they are stored in \BabelNonText, used in some special cases when no Latin encoding is explicitly set).

The foregoing rules (which are applied "at begin document") cover most of the cases. No assumption is made on characters above 127, which may not follow the LICR conventions – the goal is just to ensure most of the ASCII letters and symbols are the right ones.

5.10 Bidirectional and right-to-left text

No macros to select the writing direction are provided, either – writing direction is intrinsic to each script and therefore it is best set by the language (which can be a dummy one). Furthermore, there are in fact two right-to-left modes, depending on the language, which differ in the way 'weak' numeric characters are ordered (e.g., Arabic %123 *vs* Hebrew 123%).

NOTE According to Unicode (UAX#9): "Where available, markup should be used instead of the explicit formatting characters" (i.e., the invisible RLI, LRI, etc., characters). So, they are ignored by the babel algorithm by default (the UAX#9 does not require supporting them). However, the implicit marks LEFT-TO-RIGHT MARK (U+200E), RIGHT-TO-LEFT MARK (U+200F) and ARABIC LETTER MARK (U+061C) are recognized. See *When Unicode bidi is not enough*.

Work is currently underway to make babel compatible with the unibidi-lua package (see 26.13 ※), which supports all of formatting characters, though the rule above will still apply (even if these characters are visible with the ^^^^ notation).

WARNING The current code for **text** in luatex should be considered essentially stable, but, of course, it is not bug-free and there can be improvements in the future, because setting bidi text has many subtleties (see for example *Additional Requirements for Bidi in HTML & CSS*). A basic stable version for other engines must wait. This applies to text; there is a basic support for **graphical** elements, including the picture environment (with pict2e) and pfg/tikz. Also, indexes and the like are under study, as well as math (there are progresses in the latter, including amsmath and mathtools too, but for example gathered may fail).

An effort is being made to avoid incompatibilities in the future (this one of the reasons currently bidi must be explicitly requested as a package option, with a certain bidi model, and also the layout options described below).

WARNING If characters to be mirrored are shown without changes with luatex, try with the following line:

```
\babeladjust{bidi.mirroring=off}
```

There are some package options controlling bidi writing.

`bidi=default | basic | basic-r | bidi-l | bidi-r`

3.14 Selects the bidi algorithm to be used.

With default the bidi mechanism is just activated (by default it is not), but every change must be marked up. In pdftex this is the only option. If the RL text consists of only letters and punctuation, it will be fine in most cases, but numbers, for example, will be rendered in the wrong order.

In luatex, the preferred method is basic, which supports both L and R text. The basic-r model was a first attempt to create a bidi algorithm and provides a simple and fast method for R text in some typical cases. (They are named basic mainly because they consider only the intrinsic direction of scripts and weak directionality.)

In xetex, bidi-r and bidi-l resort to the package bidi (by Vafa Khalighi). For RL documents use the former, and for LR ones use the latter.

WARNING The package bidi patches heavily lots of macros and packages even if the RL script is not the main one, which can lead to some surprising results, so for short and simple texts (letters and punctuation) the `default` method is more often than not much preferable.

As of January 2026 the actual maintenance status of this package is unclear, so it's advisable to move to luatex, whose bidi algorithm is an integral part of babel, and therefore doesn't rely on external packages.

There are samples on GitHub, under `/required/babel/samples`. See particularly `lua-bidibasic.tex` and `lua-secenum.tex`.

EXAMPLE The following text comes from the Arabic Wikipedia (article about Arabia). Copy-pasting some text from the Wikipedia is a good way to test this feature. Remember `basic` is available in luatex only.

```
\documentclass[arabic]{article}

\usepackage[bidi=basic]{babel}

\babelfont{rm}{FreeSerif}

\begin{document}

                    وقد عرفت شبه جزيرة العرب طيلة العصر الهيليني (الاغريقي) بـ
                  Arabia أو Aravia (بالاغريقية Αραβία)، استخدم الرومان ثلاث
              بادئات بـ"Arabia" على ثلاث مناطق من شبه الجزيرة العربية، إلا أنها
                                  حقيقةً كانت أكبر مما تعرف عليه اليوم.

\end{document}
```

EXAMPLE With `bidi=basic` *both* L and R text can be mixed without explicit markup (the latter will be necessary only in some special cases where the Unicode algorithm fails), but you may want to map the font so that the correct features are in force, too. This is accomplished with an option in `\babelprovide`, as illustrated:

```
\documentclass{book}

\usepackage[english, bidi=basic]{babel}

\babelprovide[onchar=ids fonts]{arabic}

\babelfont{rm}{Crimson}
\babelfont[*arabic]{rm}{FreeSerif}

\begin{document}

Most Arabic speakers consider the two varieties to be two registers
of one language, although the two registers can be referred to in
Arabic as فصحى العصر \textit{fuṣḥā l-‘aṣr} (MSA) and
فصحى التراث \textit{fuṣḥā t-turāth} (CA).

\end{document}
```

In this example, and thanks to `onchar=ids fonts`, any Arabic letter (because the language is `arabic`) changes its font to that set for this language (here defined via `*arabic`, because Crimson does not provide Arabic letters).

NOTE Boxes are "black boxes". Numbers inside an `\hbox` (for example in a `\ref`) do not know anything about the surrounding chars. So, `\ref{A}-\ref{B}` are not rendered in the visual order A-B, but in the wrong one B-A (because the hyphen does not "see" the digits inside the `\hbox`'es). If you need `\ref` ranges, the best option is to define a

dedicated macro like this (to avoid explicit direction changes in the body; here `\texthe` must be defined to select the main language):

```
\newcommand\refrange[2]{\babelsublr{\texthe{\ref{#1}}-\texthe{\ref{#2}}}}
```

In the future a more complete method, reading recursively boxed text, may be added.

`layout=sectioning | counters | lists | contents | footnotes | captions | columns | graphics | extras | pars | nopars`

3.16 *To be expanded.* Selects which layout elements are adapted in bidi documents, including some text elements (except with options loading the `bidi` package, which provides its own mechanism to control these elements). You may use several options with a space-separated list, like `layout=counters contents sectioning` (in 3.85 ※ spaces are to be preferred over dots, which was the former syntax). This list will be expanded in future releases. Note not all options are required by all engines.

`sectioning` makes sure the sectioning macros are typeset in the main language, but with the title text in the current language (see below `\BabelPatchSection` for further details). 26.13 ※ It also isolates the text in the paragraph-like sectioning titles from the rest of the line when applying the `bidi=basic` algorithm.

`counters` required in all engines to reorder section numbers and the like (e.g., ⟨*subsection*⟩.⟨*section*⟩); required in xetex and pdftex for counters in general, as well as in luatex with `bidi=default`; required in luatex for numeric footnote marks >9 with `bidi=basic-r` (but *not* with `bidi=basic`); note, however, it can depend on the counter format.

With `counters`, `\arabic` is not only considered L text always (with `\babelsublr`, see below), but also an "isolated" block which does not interact with the surrounding chars. So, while `1.2` in R text is rendered in that order with `bidi=basic` (as a decimal number), in `\arabic{c1}.\arabic{c2}` the visual order is *c2.c1*. Of course, you may always adjust the order by changing the language, if necessary.

3.84 ※ Since `\thepage` is (indirectly) redefined, `makeindex` will reject many entries as invalid. With `counters*` babel attempts to remove the conflicting macros.

`lists` required in xetex and pdftex, but only in bidirectional (with both R and L paragraphs) documents in luatex.

WARNING `\parshape` in luatex (a TEX primitive) may produce some unexpected results, with lists horizontally misplaced if they are inside a `\vbox` (like `minipage`) and the current direction is different from the main one. A workaround in case you encounter this problem is to restore the main language before the box and then set the local one inside.

`contents` required in xetex and pdftex; in luatex toc entries are R by default if the main language is R.

`columns` required in xetex and pdftex to reverse the column order (currently only the standard two-column mode); in luatex they are R by default if the main language is R (including multicol).

`footnotes` not required in monolingual documents, but it may be useful in bidirectional documents (with both R and L paragraphs) in all engines, as well as in multilingual documents in general; you may use alternatively `\BabelFootnote` described below (what this option does exactly is also explained there).

`captions` is similar to `sectioning`, but for `\caption`; not required in monolingual documents with luatex, but may be required in xetex and pdftex in some styles (support for the latter two engines is still experimental) 3.18 .

`lrtabular, tabular` See the news page for 26.10.

`graphics` modifies the `picture` environment so that the whole figure is L but the text is R. It *does not* work with the standard `picture`, and *pict2e* is required. It attempts to do the same for pgf/tikz. Somewhat experimental. 3.32 .

`extras` is used for miscellaneous readjustments which do not fit into the previous groups. Currently redefines in luatex `\underline` and `\LaTeX2e` 3.19 .

`pars, nopars` When `layout` is used with any options in bidi documents, there are a few changes in `\@hangfrom` and also (only xetex, as it's unnecessary in luatex) in `\raggedright` and `\raggedleft`. Using `pars` applies these adjustments even without other options; conversely, `nopars` prevents these adjustments even when other options are specified 25.9 ※ .

EXAMPLE Typically, in an Arabic document you would need (luatex):

```
\usepackage[bidi=basic,
            layout=counters]{babel}
```

or (xetex):

```
\usepackage[bidi=bidi,
            layout=counters]{babel}
```

`\babelsublr{`⟨*lr-text*⟩`}`

Digits in pdftex must be marked up explicitly (unlike luatex with `bidi=basic` or `bidi=basic-r` and, usually, xetex). This command is provided to set `{`⟨*lr-text*⟩`}` in L mode if necessary. It's intended for what Unicode calls weak characters, because words are best set with the corresponding language. For this reason, there is no `rl` counterpart.

Any `\babelsublr` in *explicit* L mode is ignored. However, with `bidi=basic` and *implicit* L, it first returns to R and then switches to explicit L. To clarify this point, consider, in an R context:

```
RTL A ltr text \thechapter{} and still ltr RTL B
```

There are *three* R blocks and *two* L blocks, and the order is *RTL B and still ltr 1 ltr text RTL A*. This is by design to provide the proper behavior in the most usual cases — but if you need to use `\ref` in an L text inside R, the L text must be marked up explicitly; for example:

```
RTL A \foreignlanguage{english}{ltr text \thechapter{} and still ltr} RTL B
```

An empty `\babelsublr{}` may serve sometimes as a 'barrier' to prevent the text preceding being ordered with the following.

`\localerestoredirs`

3.86 ※ *LuaTeX*. This command resets the internal text, paragraph and body directions to those of the current locale (if different). Sometimes changing these values directly can be useful for some hacks, and this command helps in restoring the directions to the correct ones. It can be used in > arguments of array, too.

`\BabelPatchSection{`⟨*section-name*⟩`}`

Mainly for bidi text, but it can be useful in other cases. `\BabelPatchSection` and the corresponding option `layout=sectioning` takes a more logical approach (at least in many cases) because it applies the global language to the section format (including the `\chaptername` in `\chapter`), while the section text is still the current language. The latter is passed to tocs and marks, too, and with `sectioning` in `layout` they both reset the "global" language to the main one, while the text uses the "local" language.

With `layout=sectioning` all the standard sectioning commands are redefined (it also "isolates" the page number in heads, for a proper bidi behavior), but with this command you can set them individually if necessary (but note then tocs and marks are not touched).

`\BabelFootnote{`⟨*cmd*⟩`}{`⟨*local-language*⟩`}{`⟨*before*⟩`}{`⟨*after*⟩`}`

3.17 Something like:

```
\BabelFootnote{\parsfootnote}{\localename}{(}{)}
```

defines `\parsfootnote` so that `\parsfootnote{note}` is equivalent to:

```
\footnote{(\foreignlanguage{\localename}{note})}
```

but the footnote itself is typeset in the main language (to unify its direction as well as the footnote mark). In addition, `\parsfootnotetext` is defined. The option `footnotes` just does the following:

```
\BabelFootnote{\footnote}{\localename}{}{}%
\BabelFootnote{\localfootnote}{\localename}{}{}%
\BabelFootnote{\mainfootnote}{}{}{}
```

(which also redefine `\footnotetext` and define `\localfootnotetext` and `\mainfootnotetext`). If the language argument is empty, then no language is selected inside the argument of the footnote. (25.9 ※ Formerly only available in bidi documents and Unicode engines, now it's always available in all engines.)

EXAMPLE If you want to preserve directionality in footnotes and there are many footnotes entirely in English, you can define:

```
\BabelFootnote{\enfootnote}{english}{}{.}
```

It adds a period outside the English part, so that it is placed at the left in the last line. This means the dot at the end of the footnote text should be omitted.

5.11 Unicode character properties in luatex

3.32 Part of the babel job is to apply Unicode rules to some script-specific features based on some properties. Currently, they are 3, namely, direction (i.e., bidi class), mirroring glyphs, and line breaking for CJK scripts. These properties are stored in lua tables, which you can modify with the following macro (for example, to set them for glyphs in the PUA).

`\babelcharproperty{`⟨*char-code*⟩`}[`⟨*to-char-code*⟩`]{`⟨*property*⟩`}{`⟨*value*⟩`}`

3.32 Here, `{`⟨*char-code*⟩`}` is a number (with TEX syntax). With the optional argument, you can set a range of values. There are three properties (with a short name, taken from Unicode): `direction` (`bc`), `mirror` (`bmg`), `linebreak` (`lb`). The settings are global, and this command is allowed only in vertical mode (the preamble or between paragraphs).

For example:

```
\babelcharproperty{`¿}{mirror}{`?}
\babelcharproperty{`-}{direction}{l}  % or al, r, en, an, on, et, cs
\babelcharproperty{`)}{linebreak}{cl} % or id, op, cl, ns, ex, in, hy
```

Please, refer to the Unicode standard (Annex #9 and Annex #14) for the meaning of the available codes. For example, `en` is 'European number' and `id` is 'ideographic'.

3.39 ※ Another property is `locale`, which adds characters to the list used by `onchar` in `\babelprovide`, or, if the last argument is empty, removes them. The last argument is the locale name:

```
\babelcharproperty{`,}{locale}{english}
```

5.12 Tweaking some babel features

`\babeladjust{`⟨*key-value-list*⟩`}`

3.36 ※ Sometimes you might need to disable some babel features. Currently this macro

understands the following keys, with values on or off:

```
autoload.bcp47          bidi.math               layout.tabular
bcp47.toname            linebreak.sea           layout.lists
bidi.mirroring          linebreak.cjk
bidi.text               justify.arabic
```

The first four are documented elsewhere. The following are by default on, but with off can disable some features: bidi.math (only preamble) changes for math or amsmath, linebreak.sea, linebreak.cjk and justify.arabic the corresponding algorithms, layout.tabular and layout.lists changes for tabular and lists. Some of them are reverted only to some extent.

Other keys are:

```
autoload.options        prehyphenation.disable  select.encoding
autoload.bcp47.prefix   interchar.disable
autoload.bcp47.options  select.write
```

Most of them are documented elsewhere. With select.encoding=off, the encoding is not set when loading a language on the fly with pdftex (only off). prehyphenation.disable is by default nohyphenation, which means luatex prehyphenation transforms are not applied if the current hyphenation rules are nohyphenation; with off they are never disabled. interchar.disable takes the same values, but for the xetex interchar mechanism.

For example, you can set \babeladjust{bidi.text=off} if you are using an alternative algorithm or with large sections not requiring it. Use with care, because these options do not deactivate other related options (like paragraph direction with bidi.text).

6 Relation with other packages

6.1 Compatibility

There are compatibility issues with the following packages.

polyglossia	This package is not compatible at all with babel. They should never be used together, because macros are different, and even those with the same name don't behave in the same way. Also, languages are organized quite differently (babel treats all locales on an equal footing).
luabidi	Don't load this package. Use instead the babel built-in tools for bidi text.
tikz/pgf	There are some issues with shorthands, which are usually solved with \usetikzlibrary{babel}.
cleveref	Because of a long standing bug in this package, some languages can raise an error (particularly spanish and greek).
natbib	Load it before babel.
bidi	Babel relies on BIDI with xetex and bidi=bidi, which has a multitude of incompatibilities. Please, refer to its manual, and remember for short and simple texts you can rely on bidi=default instead.

6.2 Related packages

The following packages can be useful in multilingual contexts (the list is far from exhaustive):

babelbib Multilingual bibliographies.
biblatex Programmable bibliographies and citations.
bicaption Bilingual captions.
csquotes Logical markup for quotes.
hyphsubst Selects a different set of patterns for a language.
iflang Tests correctly the current language.
microtype Adjusts the typesetting according to some languages (kerning and spacing). Ligatures can be disabled. Search in its manual for `babel` and `DeclareMicrotypeBabelHook`.
mkpattern Generates hyphenation patterns.
siunitx Typesetting of numbers and physical quantities.
substitutefont Combines fonts in several encodings.
tracklang Tracks which languages have been requested.
translator An open platform for packages that need to be localized.
ucharclasses (xetex) Switches fonts when you switch from one Unicode block to another. Note it doesn't work with RTL scripts.
zhspacing Spacing for CJK documents in xetex.

6.3 Indexing

For multilingual indexing, see upmendex and xindex, currently preferred to xindy.

7 Tentative and experimental code

See the code section for `\foreignlanguage*` (a new starred version of `\foreignlanguage`).
For old and deprecated functions, see the babel site.

Options for locales loaded on the fly
3.51 ※ `\babeladjust{ autoload.options = ... }` sets the options when a language is loaded on the fly. 24.14 ※ By default, it is `import`, which defines captions, date, numerals, etc., but ignores the code in the `tex` file (for example, extended numerals in Greek). It can be set to empty.

Labels
3.48 ※ There is some work in progress for babel to deal with labels, both with the relation to captions (chapters, part), and how counters are used to define them. It is still somewhat tentative because it is far from trivial – see the babel site for further details.

8 Loading language hyphenation rules with `language.dat`

TEX and most engines based on it (pdfTEX, xetex, ε-TEX, the main exception being luatex) require hyphenation patterns to be preloaded when a format is created (e.g., LATEX, XeLATEX, pdfLATEX). babel provides a tool which has become standard in many distributions and based on a "configuration file" named `language.dat`. The exact way this file is used depends on the distribution, so please, read the documentation for the latter (note also some distributions generate the file with some tool).
3.9q With luatex, however, patterns are loaded on the fly when requested by the language (except the "0th" language, typically english, which is preloaded always). You may want to have a local `language.dat` for a particular project (for example, a book on Chemistry). The loader for lua(e)tex is slightly different as it's not based on babel, but on `etex.src`; however, babel reloads the data so that it works as expected.

8.1 Format

In that file the person who maintains a TEX environment has to record for which languages he has hyphenation patterns *and* in which files these are stored (this was because different operating systems sometimes used *very* different file-naming conventions, but this issue is

not currently so serious as before). When hyphenation exceptions are stored in a separate file this can be indicated by naming that file *after* the file with the hyphenation patterns.

The file can contain empty lines and comments, as well as lines which start with an equals (=) sign. Such a line will instruct LaTeX that the hyphenation patterns just processed have to be known under an alternative name. Here is an example:

```
% File    : language.dat
% Purpose : tell iniTeX what files with patterns to load.
english    english.hyphenations
=british

dutch      hyphen.dutch exceptions.dutch % Nederlands
german hyphen.ger
```

You may also set the font encoding that the patterns are intended for by following the language name by a colon and the encoding code. For example:

```
german:T1 hyphenT1.ger
german hyphen.ger
```

With the previous settings, if the encoding when the language is selected is `T1` then the patterns in `hyphenT1.ger` are used, but otherwise use those in `hyphen.ger` (note the encoding can be set in `\extras`⟨*language*⟩).

9 The interface between the core of babel and the language definition files

The *language definition files* (ldf) must conform to a number of conventions, because these files have to fill in the gaps left by the common code in `babel.def`, i.e., the definitions of the macros that produce texts. Also the language-switching possibility which has been built into the babel system has its implications.

The following assumptions are made:

- Some of the language-specific definitions might be used by plain TeX users, so the files have to be coded so that they can be read by both LaTeX and plain TeX. The current format can be checked by looking at the value of the macro `\fmtname`.
- The common part of the babel system redefines a number of macros and environments (defined previously in the document style) to put in the names of macros that replace the previously hard-wired texts. These macros have to be defined in the language definition files.
- The language definition files must define five macros, used to activate and deactivate the language-specific definitions. These macros are `\`⟨*language*⟩`hyphenmins`, `\captions`⟨*language*⟩, `\date`⟨*language*⟩, `\extras`⟨*language*⟩ and `\noextras`⟨*language*⟩(the last two may be left empty); where ⟨*language*⟩ is either the name of the language definition file or the name of the LaTeX option that is to be used. These macros and their functions are discussed below. You must define all or none for a language; defining, say, `\date`⟨*language*⟩ but not `\captions`⟨*language*⟩ does not raise an error but can lead to unexpected results.
- When a language definition file is loaded, it can define `\l@`⟨*language*⟩ to be a dialect of `\language0` when `\l@`⟨*language*⟩ is undefined.
- Language names must be all lowercase. If an unknown language is selected, babel will attempt setting it after lowercasing its name.
- The semantics of modifiers is not defined (on purpose). In most cases, they will just be simple separated options (e.g., `spanish`), but a language might require, say, a set of options organized as a tree with suboptions (in such a case, the recommended

separator is /). How modifiers (saved in \BabelModifiers) are handled are left to language styles; they can use \in@, loop them with \@for or load keyval, for example.

Some recommendations:

- The preferred shorthand is ", which is not used in LaTeX (quotes are entered as `` and ''). Other good choices are characters which are not used in a certain context (e.g., = in an ancient language). Note however =, <, >, : and the like can be dangerous, because they may be used as part of the syntax of some elements (numeric expressions, key/value pairs, etc.).
- Captions should not contain shorthands or encoding-dependent commands (the latter is not always possible, but should be clearly documented). They should be defined using the LICR. You may also use the new tools for encoded strings, described below.
- Avoid adding things to \noextras⟨*language*⟩ except for umlauthigh and friends, \bbl@deactivate, \bbl@(non)frenchspacing, and language-specific macros. Use always, wherever possible, \babel@save and \babel@savevariable (except if you still want to have access to the previous value). Do not reset a macro or a setting to a hardcoded value. Never. Instead save its value in \extras⟨*language*⟩.
- Do not switch scripts. If you want to make sure a set of glyphs is used, switch either the font encoding (low-level) or the language (high-level, which in turn may switch the font encoding). Usage of things like \latintext is deprecated (but not removed, for backward compatibility).
- Please, for "private" internal macros do not use the \bbl@ prefix. It is used by babel and it can lead to incompatibilities.

There are no special requirements for documenting your language files. Just provide a standalone document suited to your needs, as well as other files you think can be useful. A PDF and a "readme" are strongly recommended.

9.1 Guidelines for contributed languages

Currently, the easiest way to contribute a new language is by taking one of the 500 or so ini templates available on GitHub as a basis. Just make a pull request or download it, and then, after filling out the fields, sent it to me. Feel free to ask for help or to make feature requests.

As to ldf files, now language files are "outsourced" and are located in a separate directory (/macros/latex/contrib/babel-contrib), so that they are contributed directly to CTAN (please, do not send to me language styles just to upload them to CTAN).

Of course, placing your style files in this directory is not mandatory, but if you want to do it, here are a few guidelines.

- Do not hesitate stating on the file heads you are the author and the maintainer, if you actually are. There is no need to state the babel maintainer(s) as author(s) if they have not contributed significantly to your language files.
- Fonts are not strictly part of a language, so they are best placed in the corresponding TeX tree. This includes not only tfm, vf, ps1, otf, mf files and the like, but also fd ones.
- Font and input encodings are usually best placed in the corresponding tree, too, but sometimes they belong more naturally to the babel style. Note you may also need to define a LICR (*TLC3*, I, 757f.).
- Babel ldf files may just interface a framework, as it happens often with Oriental languages/scripts. This framework is best placed in its own directory.

The following page provides a starting point for `ldf` files: `http://www.texnia.com/incubator.html`. See also `https://latex3.github.io/babel/guides/list-of-locale-templates.html`.

If you need further assistance and technical advice in the development of language styles, I will be happy to help you. And of course, you can make any suggestion you like.

9.2 Basic macros

In the core of the babel system, several macros are defined for use in language definition files. Their purpose is to make a new language known. The first two are related to hyphenation patterns.

`\addlanguage` The macro `\addlanguage` is a non-outer version of the macro `\newlanguage`, defined in `plain.tex` version 3.x. Here "language" is used in the TEX sense of set of hyphenation patterns.

`\adddialect` The macro `\adddialect` can be used when two languages can (or must) use the same hyphenation patterns. This can also be useful for languages for which no patterns are preloaded in the format. In such cases the default behavior of the babel system is to define this language as a 'dialect' of the language for which the patterns were loaded as `\language0`. Here "language" is used in the TEX sense of set of hyphenation patterns, while "dialect", which is misnomer, refers just to a language with the same hyphenation patterns as another (there is no relation with its linguistic meaning).

`\<lang>hyphenmins` The macro \⟨*language*⟩`hyphenmins` is used to store the values of the `\lefthyphenmin` and `\righthyphenmin`. Redefine this macro to set your own values, with two numbers corresponding to these two parameters. For example:

```
\renewcommand\spanishhyphenmins{34}
```

(Assigning `\lefthyphenmin` and `\righthyphenmin` directly in `\extras`⟨*language*⟩ has no effect.) This is a low-level setting. In documents you should rely on `\babelhyphenmins`.

`\providehyphenmins` The macro `\providehyphenmins` should be used in the language definition files to set `\lefthyphenmin` and `\righthyphenmin`. This macro will check whether these parameters were provided by the hyphenation file before it takes any action. If these values have been already set, this command is ignored (currently, default pattern files do *not* set them).

`\captions`⟨*language*⟩ The macro `\captions`⟨*language*⟩ defines the macros that hold the texts to replace the original hard-wired texts.

`\date`⟨*language*⟩ The macro `\date`⟨*language*⟩ defines `\today`.

`\extras`⟨*language*⟩ The macro `\extras`⟨*language*⟩ contains all the extra definitions needed for a specific language. This macro, like the following, is a hook – you can add things to it, but it must not be used directly.

`\noextras`⟨*language*⟩ Because we want to let the user switch between languages, but we do not know what state TEX might be in after the execution of `\extras`⟨*language*⟩, a macro that brings TEX into a predefined state is needed. It will be no surprise that the name of this macro is `\noextras`⟨*language*⟩.

`\bbl@declare@ttribute` This is a command to be used in the language definition files for declaring a language attribute. It takes three arguments: the name of the language, the attribute to be defined, and the code to be executed when the attribute is to be used.

`\main@language` To postpone the activation of the definitions needed for a language until the beginning of a document, all language definition files should use `\main@language` instead of `\selectlanguage`. This will just store the name of the language, and the proper language will be activated at the start of the document.

`\ProvidesLanguage` The macro `\ProvidesLanguage` should be used to identify the language definition files. Its syntax is similar to the syntax of the LATEX command `\ProvidesPackage`.

`\LdfInit` The macro `\LdfInit` performs a couple of standard checks that must be made at the beginning of a language definition file, such as checking the category code of the @-sign, preventing the `ldf` file from being processed twice, etc.

`\ldf@quit` The macro `\ldf@quit` does work needed if an `ldf` file was processed earlier. This includes resetting the category code of the @-sign, preparing the language to be activated at `\begin{document}` time, and ending the input stream.

`\ldf@finish` The macro `\ldf@finish` does work needed at the end of each `ldf` file. This includes resetting the category code of the @-sign, loading a local configuration file, and preparing the language to be activated at `\begin{document}` time.

`\loadlocalcfg` After processing a language definition file, LaTeX can be instructed to load a local configuration file. This file can, for instance, be used to add strings to `\captions`⟨*language*⟩ to support local document classes. The user will be informed that this configuration file has been loaded. This macro is called by `\ldf@finish`.

9.3 Skeleton

Here is the basic structure of an `ldf` file, with a language, a dialect and an attribute. Strings are best defined using the method explained in sec. 9.8 (babel 3.9 and later).

```
\ProvidesLanguage{<language>}
     [2016/04/23 v0.0 <Language> support from the babel system]
\LdfInit{<language>}{captions<language>}

\ifx\undefined\l@<language>
  \@nopatterns{<Language>}
  \adddialect\l@<language>0
\fi

\adddialect\l@<dialect>\l@<language>

\bbl@declare@ttribute{<language>}{<attrib>}{%
  \expandafter\addto\expandafter\extras<language>
  \expandafter{\extras<attrib><language>}%
  \let\captions<language>\captions<attrib><language>}

\providehyphenmins{<language>}{\tw@\thr@@}

\StartBabelCommands*{<language>}{captions}
\SetString\chaptername{<chapter name>}
% More strings

\StartBabelCommands*{<language>}{date}
\SetString\monthiname{<name of first month>}
% More strings

\StartBabelCommands*{<dialect>}{captions}
\SetString\chaptername{<chapter name>}
% More strings

\StartBabelCommands*{<dialect>}{date}
\SetString\monthiname{<name of first month>}
% More strings

\EndBabelCommands

\addto\extras<language>{}
\addto\noextras<language>{}
\let\extras<dialect>\extras<language>
\let\noextras<dialect>\noextras<language>

\ldf@finish{<language>}
```

NOTE If for some reason you want to load a package in your style, you should be aware it cannot be done directly in the `ldf` file, but it can be delayed with `\AtEndOfPackage`.

Macros from external packages can be used *inside* definitions in the ldf itself (for example, `\extras`⟨*language*⟩), but if executed directly, the code must be placed inside `\AtEndOfPackage`. A trivial example illustrating these points is:

```
\AtEndOfPackage{%
  \RequirePackage{dingbat}%       Delay package
  \savebox{\myeye}{\eye}}%        And direct usage
\newsavebox{\myeye}
\newcommand\myanchor{\anchor}%    But OK inside command
```

9.4 Support for active characters

In quite a number of language definition files, active characters are introduced. To facilitate this, some support macros are provided.

`\initiate@active@char` Used in language definition files to instruct LaTeX to give a character the category code 'active'. When a character has been made active it will remain that way until the end of the document. Its definition may vary.

`\bbl@activate` `\bbl@deactivate` Used to change the way an active character expands. `\bbl@activate` 'switches on' the active behavior of the character. `\bbl@deactivate` lets the active character expand to its former (mostly) non-active self.

`\declare@shorthand` Used to define the various shorthands. It takes three arguments: the name for the collection of shorthands this definition belongs to; the character (sequence) that makes up the shorthand, i.e., ~ or `"a`; and the code to be executed when the shorthand is encountered. (It does *not* raise an error if the shorthand character has not been "initiated".)

`\bbl@add@special` The TeXbook states: "Plain TeX includes a macro called `\dospecials` that is essentially a set macro, representing the set of all characters that have a special category code." [4, p. 380] It is used to set text 'verbatim'. To make this work if more characters get a special category code, you have to add this character to the macro `\dospecials`. LaTeX adds another macro called `\@sanitize` representing the same character set, but without the curly braces. The macros `\bbl@add@special{`*char*`}` add the character ⟨*char*⟩ to these two sets.

`\@safe@activestrue` `\@safe@activesfalse` Enables and disables the "safe" mode. It is a tool for package and class authors. See the description below.

9.5 Support for saving macro definitions

Language definition files may want to *re*define macros that already exist. Therefore a mechanism for saving (and restoring) the original definition of those macros is provided. We provide two macros for this.[7]

`\babel@save` To save the current meaning of any control sequence, the macro `\babel@save` is provided. It takes one argument, ⟨*csname*⟩, the control sequence for which the meaning has to be saved.

`\babel@savevariable` A second macro is provided to save the current value of a variable. In this context, anything that is allowed after the `\the` primitive is considered to be a variable. The macro takes one argument, the ⟨*variable*⟩.

The effect of the preceding macros is to append a piece of code to the current definition of `\originalTeX`. When `\originalTeX` is expanded, this code restores the previous definition of the control sequence or the previous value of the variable.

9.6 Support for extending macros

`\addto` The macro `\addto{`⟨*control sequence*⟩`}{`⟨*TEX code*⟩`}` can be used to extend the definition of a macro. The macro need not be defined (i.e., it can be undefined or `\relax`). This macro can, for instance, be used in adding instructions to a macro like `\extrasenglish`.

[7] This mechanism was introduced by Bernd Raichle.

Be careful when using this macro, because depending on the case the assignment can be either global (usually) or local (sometimes). That does not seem very consistent, but this behavior is preserved for backward compatibility. If you are using `etoolbox`, by Philipp Lehman, consider using the tools provided by this package instead of `\addto`.

9.7 Macros common to a number of languages

`\bbl@allowhyphens` In several languages compound words are used. This means that when TEX has to hyphenate such a compound word, it only does so at the '-' that is used in such words. To allow hyphenation in the rest of such a compound word, the macro `\bbl@allowhyphens` can be used.

`\allowhyphens` Same as `\bbl@allowhyphens`, but does nothing if the encoding is `T1`. It is intended mainly for characters provided as real glyphs by this encoding but constructed with `\accent` in `OT1`.

Note the previous command (`\bbl@allowhyphens`) has different applications (hyphens and discretionaries) than this one (composite chars). Note also prior to version 3.7, `\allowhyphens` had the behavior of `\bbl@allowhyphens`.

`\set@low@box` For some languages, quotes need to be lowered to the baseline. For this purpose the macro `\set@low@box` is available. It takes one argument and puts that argument in an `\hbox`, at the baseline. The result is available in `\box0` for further processing.

`\save@sf@q` Sometimes it is necessary to preserve the `\spacefactor`. For this purpose the macro `\save@sf@q` is available. It takes one argument, saves the current spacefactor, executes the argument, and restores the spacefactor.

`\bbl@frenchspacing` `\bbl@nonfrenchspacing` Those commands can be used to properly switch French spacing on and off.

NOTE With `ldf` files, babel does not take into account `\normalsfcodes` and (non-)French spacing is not always properly (un)set by languages However, problems are unlikely to happen and therefore this part remains untouched. With `ini` files, this issue has been addressed.

9.8 Encoding-dependent strings

3.9a Babel 3.9 provides a way of defining strings in several encodings, intended mainly for luatex and xetex, although the old way of defining/switching strings still works and it's used by default.

It consists in a series of blocks started with `\StartBabelCommands`. The last block is closed with `\EndBabelCommands`. Each block is a single group (i.e., local declarations apply until the next `\StartBabelCommands` or `\EndBabelCommands`). An `ldf` may contain several series of this kind.

Thanks to this new feature, string values and string language switching are not mixed any more. Furthermore, strings do not need to be wrapped with formatting commands (e.g., to select the writing direction) because babel takes care of it automatically. (See also `\setlocalecaption`.)

`\StartBabelCommands{⟨language-list⟩}{⟨category⟩}[⟨selector⟩]`

The ⟨*language-list*⟩ specifies which languages the block is intended for. A block is taken into account only if the `\CurrentOption` is listed here. Alternatively, you can define `\BabelLanguages` to a comma-separated list of languages to be defined (if undefined, `\StartBabelCommands` sets it to `\CurrentOption`). You may write `\CurrentOption` as the language, but this is discouraged – an explicit name (or names) is much better and clearer.

A "selector" selects a group of definition that are to be used, optionally followed by extra info about the encodings to be used. The name `unicode` must be used for luatex and xetex. Without a selector, the LICR representation (i.e., with macros like `\~{n}` instead of ñ) is assumed.

If a string is set several times (because several blocks are read), the first one takes precedence (i.e., it works much like `\providecommand`).

Encoding info is `charset=` followed by a charset, which, if given, sets how the strings should be translated to the internal representation used by the engine, typically `utf8`, which is the only value supported currently (default is no translations). Note `charset` is applied by luatex and xetex when reading the file, not when the macro or string is used in the document.

A list of font encodings which the strings are expected to work with can be given after `fontenc=` (separated with spaces, if two or more) – recommended, but not mandatory, although blocks without this key are not taken into account if you have requested `strings=encoded`.

Blocks without a selector are read always. They provide fallback values, and therefore they must be the last ones; they should be provided always if possible and all strings should be defined somehow inside it; they can be the only blocks (mainly LGC scripts using the LICR).

The ⟨*category*⟩ is either `captions`, `date` or `extras`. You must stick to these three categories, even if no error is raised when using other names.[8] It may be empty, too, but in such a case using `\SetString` is an error.

```
\StartBabelCommands{language}{captions}
  [unicode, fontenc=TU, charset=utf8]
\SetString{\chaptername}{utf8-string}

\StartBabelCommands{language}{captions}
\SetString{\chaptername}{ascii-maybe-LICR-string}

\EndBabelCommands
```

A real example can be:

```
\StartBabelCommands{austrian}{date}
  [unicode, fontenc=TU, charset=utf8]
  \SetString\monthiname{Jänner}

\StartBabelCommands{german,austrian}{date}
  [unicode, fontenc=TU, charset=utf8]
  \SetString\monthiiiname{März}

\StartBabelCommands{austrian}{date}
  \SetString\monthiname{J\"{a}nner}

\StartBabelCommands{german}{date}
  \SetString\monthiname{Januar}

\StartBabelCommands{german,austrian}{date}
  \SetString\monthiiname{Februar}
  \SetString\monthiiiname{M\"{a}rz}
  \SetString\monthivname{April}
  \SetString\monthvname{Mai}
  \SetString\monthviname{Juni}
  \SetString\monthviiname{Juli}
  \SetString\monthviiiname{August}
  \SetString\monthixname{September}
  \SetString\monthxname{Oktober}
  \SetString\monthxiname{November}
  \SetString\monthxiiname{Dezenber}
  \SetString\today{\number\day.~%
    \csname month\romannumeral\month name\endcsname\space
    \number\year}

\StartBabelCommands{german,austrian}{captions}
```

[8] In future releases further categories may be added.

```
  \SetString\prefacename{Vorwort}
  [etc.]

\EndBabelCommands
```

When used in `ldf` files, previous values of \⟨*category*⟩⟨*language*⟩ are overridden, which means the old way to define strings still works and used by default (to be precise, is first set to undefined and then strings are added). However, when used in the preamble or in a package, new settings are added to the previous ones, if the language exists (in the babel sense, i.e., if `\date`⟨*language*⟩ exists).

NOTE The package option `strings` introduced in version 3.9 (around 2013) when Unicode engines were still of marginal use, is now deprecated.

NOTE Captions and other strings defined in `ini` files (in other words, when a locale is loaded with `\babelprovide`) are internally set with the help of these macros.

`\StartBabelCommands` `*` `{`⟨*language-list*⟩`}{`⟨*category*⟩`}[`⟨*selector*⟩`]`

The starred version just forces `strings` to take a value – if not set as package option (which is now deprecated), then the default for the engine is used. This is not done by default to prevent backward incompatibilities, but if you are creating a new language this version is better. It's up to the maintainers of the current languages to decide if using it is appropriate.

`\EndBabelCommands`

Marks the end of the series of blocks.

`\AfterBabelCommands{`⟨*code*⟩`}`

The code is delayed and executed at the global scope just after `\EndBabelCommands`.

`\SetString{`⟨*macro-name*⟩`}{`⟨*string*⟩`}`

Adds ⟨*macro-name*⟩ to the current category, and defines globally ⟨*lang-macro-name*⟩ to ⟨*code*⟩ (after applying the transformation corresponding to the current charset or defined with the hook `stringprocess`).

Use this command to define strings, without including any "logic" if possible, which should be a separate macro. See the example above for the date.

`\SetStringLoop{`⟨*macro-name*⟩`}{`⟨*string-list*⟩`}`

A convenient way to define several ordered names at once. For example, to define `\abmoniname`, `\abmoniiname`, etc. (and similarly with `abday`):

```
\SetStringLoop{abmon#1name}{en,fb,mr,ab,my,jn,jl,ag,sp,oc,nv,dc}
\SetStringLoop{abday#1name}{lu,ma,mi,ju,vi,sa,do}
```

`#1` is replaced by the roman numeral.

`\SetHyphenMap{`⟨*to-lower-macros*⟩`}`

3.9g Case mapping for hyphenation is handled with `\SetHyphenMap` and controlled with the package option `hyphenmap`.

There are three helper macros to be used inside `\SetHyphenMap`:

- `\BabelLower{`⟨*uccode*⟩`}{`⟨*lccode*⟩`}` is similar to `\lccode` but it's ignored if the char has been set and saves the original lccode to restore it when switching the language (except with `hyphenmap=first`).
- `\BabelLowerMM{`⟨*uccode-from*⟩`}{`⟨*uccode-to*⟩`}{`⟨*step*⟩`}{`⟨*lccode-from*⟩`}` loops though the given uppercase codes, using the step, and assigns them the lccode, which is also increased (`MM` stands for *many-to-many*).

- `\BabelLowerMO{`*uccode-from*`}{`*uccode-to*`}{`*step*`}{`*lccode*`}` loops though the given uppercase codes, using the step, and assigns them the lccode, which is fixed (`MO` stands for *many-to-one*).

An example is (which is redundant, because these assignments are done by both luatex and xetex):

```
\SetHyphenMap{\BabelLowerMM{"100}{"11F}{2}{"101}}
```

NOTE This macro is not intended to fix wrong mappings done by Unicode (which are the default in both luatex and xetex) – if an assignment is wrong, fix it directly.

10 Acknowledgements

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References

[1] Huda Smitshuijzen Abifares, *Arabic Typography*, Saqi, 2001.

[2] Johannes Braams, Victor Eijkhout and Nico Poppelier, *The development of national LaTeX styles*, *TUGboat* 10 (1989) #3, pp. 401–406.

[3] Yannis Haralambous, *Fonts & Encodings*, O'Reilly, 2007.

[4] Donald E. Knuth, *The TeXbook*, Addison-Wesley, 1986.

[5] Jukka K. Korpela, *Unicode Explained*, O'Reilly, 2006.

[6] Leslie Lamport, *LaTeX, A document preparation System*, Addison-Wesley, 1986.

[7] Leslie Lamport, in: TeXhax Digest, Volume 89, #13, 17 February 1989.

[8] Ken Lunde, *CJKV Information Processing*, O'Reilly, 2nd ed., 2009.

[9] Edward M. Reingold and Nachum Dershowitz, *Calendrical Calculations: The Ultimate Edition*, Cambridge University Press, 2018

[10] Hubert Partl, *German TeX*, *TUGboat* 9 (1988) #1, pp. 70–72.

[11] Joachim Schrod, *International LaTeX is ready to use*, *TUGboat* 11 (1990) #1, pp. 87–90.

[12] Apostolos Syropoulos, Antonis Tsolomitis and Nick Sofroniu, *Digital typography using LaTeX*, Springer, 2002, pp. 301–373.

[13] K.F. Treebus. *Tekstwijzer, een gids voor het grafisch verwerken van tekst*, SDU Uitgeverij ('s-Gravenhage, 1988).